Generating integrable systems from an autonomous ordinary differential equation

Graham Hesketh
gdh1e10@gmail.com

October 18, 2026

Abstract

The freedom to realise special-function dynamics within integrable Kummer-type Hamiltonian systems remains underexplored. We investigate this freedom through examples constructed from prescribed autonomous scalar equations, using the established reduction and mode-reconstruction framework. A local holomorphic equation $\rho' = f(\rho)$ determines a family of canonical Hamiltonians on a selected energy level, subject to regularity and compatible branch choices. We introduce the category of log-augmented polynomial systems, whose polynomial mode equations retain logarithmic terms in their scalar reductions, and illustrate it with examples including the Lotka–Volterra model and a parity-time-symmetric dimer. For rational generating equations, we prove a correspondence with extended Lambert–W-type inverse laws and establish elementary evaluation of rational reconstruction quadratures, giving local solvability results for the associated coupled systems. Further constructions use inverse incomplete beta and inverse spherical Bessel functions and the modular λ function. The modular example has explicit theta-constant mode solutions and a direct map to the Darboux–Halphen system.

1 Introduction

Integrable systems of coupled modes provide a meeting point between nonlinear optics, classical mechanics, and special-function theory. In nonlinear optics, resonant wave mixing and nonlinear directional couplers furnish familiar examples [1, 2, 3, 4, 5, 6]. Conserved combinations of modal powers reduce these systems to a scalar autonomous differential equation. Cubic and quartic polynomial reductions lead to elliptic functions, while the individual complex mode amplitudes require additional reconstruction. Explicit solutions of optical coupled-mode equations reveal connections with classical mechanical systems and identities among elliptic functions, motivating the search for a common constructive framework [6, 7, 8, 9].

Under suitable parameter choices and initial conditions, these coupled-mode systems also admit projections to classical mechanical systems, including the Lagrange top and the Duffing oscillator; the Euler top appears as a special reduction, as discussed in the examples below [10, 11].

The construction of integrable Hamiltonian systems has several established approaches. Lax representations relate the equations of motion to spectral invariants, while separation relations can be used to define commuting Hamiltonians and their separated equations. Stäckel transforms and coupling-constant metamorphosis provide ways to obtain related integrable systems from known ones [12, 13, 14]. These approaches offer a broad context for the present work, which concerns a particular family of resonantly coupled modes and uses a scalar autonomous equation as the starting point for its construction.

The closest precedents are the resonant Hamiltonians studied by Kummer and their subsequent generalisations. Kummer developed reduction and integration for Hamiltonians with polynomial dependence on oscillator powers and constant resonant coupling [15]. Related work

on integrable multiboson systems and intensity-dependent conversion allowed functions of mode occupations to enter both the diagonal and interaction terms, with explicit constructions linked to orthogonal polynomials [16, 17]. Odzijewicz and Wawreniuk formulated a classical and quantum Kummer-shape framework with smooth power-dependent functions in place of the polynomial and constant choices. Their examples include a non-polynomial classical Hamiltonian associated with a deformed oscillator algebra [18]. Their studies of three- and four-wave mixing further connect this framework with nonlinear optics [19, 20].

Our recent work extends the explicit solution theory beyond the genus-one reductions of the familiar optical systems. When nonlinear interactions produce a higher-degree scalar polynomial whose normalized curve has genus greater than one, a more general class of polynomial Kummer-type systems is accessible through hyperelliptic sigma functions and Abelian integrals [21]. That work constructs local solutions for smooth odd-degree curves under the stated degree and regularity conditions, together with reconstruction of the individual modes. Section 5.1 places these systems within the present framework.

We believe that the freedom to realise a wider range of special functions in Kummer-type systems remains underexplored. The established freedom in the coefficient functions makes it possible to explore scalar dynamics beyond the familiar elliptic examples. Polynomial mode equations can themselves conceal such behaviour: logarithmic terms in a conserved Hamiltonian may disappear from the equations of motion after the conserved value is used, yet remain in the reduced scalar equation. The classical two-species Lotka–Volterra predator–prey system is a familiar example: its polynomial equations possess a logarithmic first integral, and the scalar reduction considered below retains logarithmic terms [22]. Polynomial coupled equations therefore need not lead to an elliptic or hyperelliptic reduction. Conversely, allowing non-polynomial Hamiltonians provides a way to embed prescribed scalar dynamics and special functions into coupled canonical systems. These observations motivate treating the scalar equation as a means both of constructing systems and of comparing their dynamics.

1.1 Our contribution

Our focus is to explore this special-function freedom through explicit examples and solvability results. We use a differential-equation-first construction for canonical Hamiltonians of the form

$$\begin{aligned} &H = (U+V)P(x_1,\dots,x_N) - Q(x_1,\dots,x_N),\\ &x_j = u_j v_j, \qquad U = \prod_{j=1}^{N} u_j^{\beta_j}, \qquad V = \prod_{j=1}^{N} v_j^{\beta_j}. \end{aligned} \tag{1.1}$$

Here $u_j(z)$ and $v_j(z)$ are canonical mode variables, and the exponents β_j are integers with $\sum_j \beta_j \neq 0$; the optical and polynomial examples use positive integer weights. The functions P and Q depend only on the modal products. Up to sign and conjugacy conventions, they play the roles of the coupling and power-dependent functions in the broader Kummer framework. We treat u_j and v_j as independent canonical variables, without requiring them to be complex conjugates. The underlying resonant integrability mechanism and the admission of smooth coefficient functions are established features of the preceding literature; our purpose is to use that freedom systematically to connect prescribed scalar equations with explicit coupled-mode systems.

The intermodal power laws give

$$x_j(z) = \gamma_j - \beta_j \rho(z), \tag{1.2}$$

where the γ_j are constants fixed by the conserved quantities. On the level $H = 0$, the remaining power evolution satisfies

$$[\rho'(z)]^2 = Q(x(\rho))^2 - 4P(x(\rho))^2 \prod_{j=1}^{N} (\gamma_j - \beta_j \rho)^{\beta_j}, \tag{1.3}$$

and the individual modes are recovered by quadratures. Reading this relation in reverse gives the construction: a prescribed autonomous equation $\rho' = f(\rho)$ determines a relation between P and Q, leaving functional freedom to choose the mode equations. The squared equation requires a compatible local branch to recover the chosen sign of $\rho' = f(\rho)$. The target dynamics are realised on the specified Hamiltonian level and conserved-power relations; the construction is local, with regularity and branch conditions stated below.

One may also begin with a sufficiently regular locally invertible function $\rho(z)$. If s is its local inverse, then $f(\rho) = 1/s'(\rho)$ defines the autonomous equation wherever the inverse is regular. Thus special functions can supply generating equations as well as solve reductions obtained from existing systems. The construction works in both directions: it produces canonical systems from a chosen scalar equation and identifies common scalar dynamics underlying apparently different coupled systems. The reduction and its product relations are recalled below in a form suited to this inverse construction.

We recap the hierarchy of integrable continuous-wave nonlinear-optical mode systems and summarise its projections and reductions to classical mechanics in Section 5. This overview establishes the common structure that motivates the construction; detailed solutions and derivations of those correspondences are left to the cited literature. The main aim here is to move beyond that familiar elliptic hierarchy by using prescribed scalar dynamics to construct and organise a broader family of canonical systems.

The contributions are organised around three aspects of this freedom. First, we introduce the category of *log-augmented polynomial systems*: polynomial mode equations whose scalar reductions retain logarithmic terms and are therefore generally nonalgebraic. This category brings together examples including the Lotka–Volterra model and a parity-time-symmetric dimer.

Second, we prove that rational generating equations correspond to extended Lambert–W-type inverse laws, and establish elementary evaluation of rational reconstruction quadratures. These results describe local solvability of the associated rational coupled systems, with polynomial equations obtainable by a regular change of parameter on the selected level.

Third, we give further explicit constructions using inverse special functions, including a one-mode Fermat family expressed through incomplete beta inversion and a system generated by an inverse spherical Bessel function. A further example generated by the modular λ function admits theta-constant mode solutions and a direct map to the Darboux–Halphen system. Together, these examples demonstrate how the established functional freedom can be used to construct and organise solvable canonical systems, including examples without an immediate physical interpretation.

The paper first develops the general reduction and the inverse construction, followed by a formulation in terms of total complex modal phase. The remaining sections examine elliptic and higher-degree examples, log-augmented polynomial systems, rational generating equations and generalised Lambert–W-type inverse laws, and the two non-polynomial special-function constructions.

2 The general Hamiltonian system

In this section we recall the established reduction and reconstruction mechanism and formulate the product identities as the Main Theorem used in the subsequent construction. The notation is motivated by nonlinear optics and elliptic function theory. Unless otherwise stated, the complex constructions are local: P, Q are holomorphic on an open domain $\Omega \subset \mathbb{C}^N$, and the modes

are holomorphic on a simply connected open domain $D \subset \mathbb{C}$ with $x(D) \subset \Omega$. Meromorphic coefficients are restricted to domains avoiding their poles; roots, logarithms, and inverse functions are taken on compatible local branches, and every displayed division is restricted to a nonzero denominator. Real restrictions may be taken where these expressions are defined. These local statements assert neither global single-valuedness nor continuation through singularities.

Let $N \in \mathbb{Z}^+$ and consider canonical pairs $u_j, v_j : D \to \mathbb{C}$ $(j = 1, \ldots, N)$, with $v_j(z)$ the Hamiltonian conjugate of $u_j(z)$. In analogy with nonlinear optics we refer to u_j, v_j as modes and the product $u_j v_j$ as modal power. The modes are functions of a single independent variable z that might represent propagation length in nonlinear optics or time in quantum mechanics. For $P, Q : \Omega \to \mathbb{C}$ as above and integers $\beta_m \in \mathbb{Z}$ $(m = 1, \ldots, N)$ with $\sum_m \beta_m \neq 0$, define the canonical Hamiltonian:

$$H\left(u_1, \ldots, u_N, v_1, \ldots, v_N\right) = \left(\prod_{m=1}^{N} u_m^{\beta_m} + \prod_{m=1}^{N} v_m^{\beta_m}\right) P(u_1 v_1, \ldots, u_N v_N) - Q(u_1 v_1, \ldots, u_N v_N) \tag{2.1}$$

where if $\beta_j > 1$ it indicates degeneracy in the system such as that which appears in degenerate four-wave mixing in nonlinear optics. Once we have obtained the equations of motion from the Hamiltonian we then work on the zero energy level $H = 0$. This is not restrictive since any constant energy value can be absorbed into Q as an additive constant. Thus we have:

$$\left(\prod_{m=1}^{N} u_m^{\beta_m} + \prod_{m=1}^{N} v_m^{\beta_m}\right) P = Q. \tag{2.2}$$

Let us introduce the notation $x_j = u_j v_j$ such that:

$$P(u_1 v_1, \ldots, u_N v_N) = P(x_1, \ldots, x_N), \tag{2.3}$$

$$\partial_{v_j} P = u_j \partial_{x_j} P = u_j \left(\partial_{x_j} P\right)|_{x_j = u_j v_j}, \tag{2.4}$$

etc., then we can write the equations of motion obtained via Hamilton's equations as:

$$\begin{aligned}
\frac{du_j}{dz} &= \partial_{v_j} H, \\
&= \left\{\left(\prod_{m=1}^{N} u_m^{\beta_m} + \prod_{m=1}^{N} v_m^{\beta_m}\right) \partial_{x_j} P - \partial_{x_j} Q\right\} u_j + \frac{\beta_j P}{v_j} \prod_{m=1}^{N} v_m^{\beta_m}, \\
&= \left\{\frac{Q \partial_{x_j} P}{P} - \partial_{x_j} Q\right\} u_j + \frac{\beta_j P}{v_j} \prod_{m=1}^{N} v_m^{\beta_m},
\end{aligned}$$

$$\begin{aligned}
\frac{dv_j}{dz} &= -\partial_{u_j} H, \\
&= -\left\{\left(\prod_{m=1}^{N} u_m^{\beta_m} + \prod_{m=1}^{N} v_m^{\beta_m}\right) \partial_{x_j} P - \partial_{x_j} Q\right\} v_j - \frac{\beta_j P}{u_j} \prod_{m=1}^{N} u_m^{\beta_m}, \\
&= -\left\{\frac{Q \partial_{x_j} P}{P} - \partial_{x_j} Q\right\} v_j - \frac{\beta_j P}{u_j} \prod_{m=1}^{N} u_m^{\beta_m},
\end{aligned} \tag{2.5}$$

where the terms inside the curly brackets depend only on the modal powers $u_j v_j$ and thus, in analogy with nonlinear optics, could be viewed as phase modulation effects, whereas the final product terms deliver wave mixing among modes. It follows from (2.5) that modal power evolves according to:

$$\frac{d\left(u_j v_j\right)}{dz} = -\left(\prod_{m=1}^{N} u_m^{\beta_m} - \prod_{m=1}^{N} v_m^{\beta_m}\right) \beta_j P, \tag{2.6}$$

and thus we have intermodal power conservation laws, which express the conserved weighted power combinations known in optics as Manley–Rowe relations [1]:

$$\beta_k \frac{d\,(u_j v_j)}{dz} - \beta_j \frac{d\,(u_k v_k)}{dz} = 0. \tag{2.7}$$

Let us now define the function $\rho(z) : \mathbb{C} \to \mathbb{C}$ as the negative total modal power over the sum of the degeneracy numbers β_j, which if all $\beta_j = 1$ would be the negative mean modal power:

$$\rho(z) = -\frac{\sum_{j=1}^{N} u_j v_j}{\sum_{j=1}^{N} \beta_j}, \tag{2.8}$$

The assumption $\sum_j \beta_j \neq 0$ makes this normalization well defined. It follows that:

$$\rho' = \frac{d\rho}{dz} = \left(\prod_{m=1}^{N} u_m^{\beta_m} - \prod_{m=1}^{N} v_m^{\beta_m} \right) P, \tag{2.9}$$

$$\frac{d\,(u_j v_j)}{dz} = -\beta_j \rho', \tag{2.10}$$

$$u_j v_j = \gamma_j - \beta_j \rho, \tag{2.11}$$

$$\sum_{j=1}^{N} \gamma_j = 0 \tag{2.12}$$

where the integration constants γ_j sum to zero by the definition of $\rho(z)$ in (2.8). We thus see that all modal powers can be expressed in terms of a single z-dependent function $\rho(z)$ and therefore Q and P can be expressed as functions of $\rho(z)$. Combining (2.2), (2.6), and (2.10) we obtain the important relations:

$$\begin{aligned} \prod_{m=1}^{N} u_m^{\beta_m} &= \frac{Q + \rho'}{2P}, \\ \prod_{m=1}^{N} v_m^{\beta_m} &= \frac{Q - \rho'}{2P}. \end{aligned} \tag{2.13}$$

Section 5 places several nonlinear-optical models within this framework. In our previous work on four-wave mixing and coherent couplers, we connected the product relations (2.13) with the Frobenius–Stickelberger determinant identity for elliptic functions [8, 9]. In the companion work on hyperelliptic extensions, we associate the corresponding relations with Ônishi's hyperelliptic determinant identities and give them a Schur-complement interpretation [21, 23]. These connections suggest that the product relations encode structure beyond their immediate role in reducing the equations; the detailed determinant constructions are developed in those works. Here we use (2.13) to recast the equations of motion in (2.5) as:

$$\begin{aligned} \frac{d \log u_j}{dz} &= \left\{ \frac{Q \partial_{x_j} P}{P} - \partial_{x_j} Q \right\} + \frac{\beta_j\,(Q - \rho')}{2(\gamma_j - \beta_j \rho)}, \\ \frac{d \log v_j}{dz} &= -\left\{ \frac{Q \partial_{x_j} P}{P} - \partial_{x_j} Q \right\} - \frac{\beta_j\,(Q + \rho')}{2(\gamma_j - \beta_j \rho)}, \end{aligned} \tag{2.14}$$

thus demonstrating that the logarithmic derivatives of u_j, v_j are separable and can be expressed in terms of the chosen Q, P and ρ, ρ'.

We now return to the defining equation for ρ'. Observe the simple but important algebraic identity:

$$\left(\prod_{m=1}^{N} u_m^{\beta_m} + \prod_{m=1}^{N} v_m^{\beta_m} \right)^2 - \left(\prod_{m=1}^{N} u_m^{\beta_m} - \prod_{m=1}^{N} v_m^{\beta_m} \right)^2 = 4 \prod_{m=1}^{N} (u_m v_m)^{\beta_m} \tag{2.15}$$

This important identity is what delivers separability for the ρ' equation as we note that the equation for ρ' contains the difference of products over u_j and v_j and the Hamiltonian contains their sum. It thus follows either by squaring (2.9) and using (2.2) and (2.15), or equivalently, from multiplying the two expressions in (2.13), that:

$$\begin{aligned}\rho'^2 &= Q^2 - 4P^2 \prod_{m=1}^{N} (u_m v_m)^{\beta_m}\,,\\ \rho'^2 &= Q^2 - 4P^2 \prod_{m=1}^{N} (\gamma_m - \beta_m \rho)^{\beta_m}\,,\end{aligned} \tag{2.16}$$

and thus the differential equation for ρ is separable.

Appendix A verifies Liouville integrability on the regular locus where the conserved quantities are functionally independent.

2.1 Theorem form without Hamiltonian mechanics

The separation and reconstruction mechanism underlying the following statement is established in the Kummer literature. Kummer treated constant resonant coupling (P constant in our notation) and polynomial power dependence (Q polynomial), subject to the degree restrictions of his setting [15]. Odzijewicz and Wawreniuk allowed smooth power-dependent coupling and diagonal functions, corresponding to P and Q, and obtained the separated scalar equation and reconstruction quadratures [18, Section 2, especially Eq. (2.31)]. Their formulation uses conjugate oscillator variables and a positive coupling function; here we work locally with independent complex canonical variables and holomorphic coefficients.

We call the following statement the Main Theorem because it supplies the identities used throughout our construction, without claiming the underlying separability as a new result. We give a direct proof from the differential equations, independently of Hamiltonian conservation, making the local branch and initial-data compatibility conditions explicit. Thus we work in reverse from the equations for ρ, u_j and v_j to the identities derived in Section 2.

Theorem 2.1 (Main Theorem)**.** *Let $\Omega \subset \mathbb{C}^N$ and $D \subset \mathbb{C}$ be open, with D simply connected. Let P, Q be holomorphic on Ω, and let u_j, v_j, ρ be holomorphic on D, with $u_j v_j \neq 0$, $x(D) \subset \Omega$, and $P(x(z)) \neq 0$ throughout D. Let $\beta_j \in \mathbb{Z}$ satisfy $\sum_j \beta_j \neq 0$, and define*

$$x_j(z) = u_j(z) v_j(z), \tag{2.17}$$

$$p(z) = P\left(x_1, \ldots, x_N\right), \tag{2.18}$$

$$q(z) = Q\left(x_1, \ldots, x_N\right), \tag{2.19}$$

$$r(z) = \prod_{m=1}^{N} x_m^{\beta_m}. \tag{2.20}$$

Let prime denote differentiation with respect to z, assume $\rho'(z) \neq 0$ on D, and let ρ satisfy

$$\rho'^2 = q^2 - 4p^2 r. \tag{2.21}$$

Let u_j solve:

$$\left(\log\left(u_j\right)\right)' = \frac{\beta_j Q}{2x_j} - \partial_{x_j} Q + \frac{Q \partial_{x_j} P}{P} + \frac{\left(\log\left(x_j\right)\right)'}{2}, \tag{2.22}$$

where P, Q and their partial derivatives are evaluated at $(x_1(z), \ldots, x_N(z))$. Define v_j such that for $\gamma_j \in \mathbb{C} \quad (j = 1, \ldots, N) : \sum_{j=1}^{N} \gamma_j = 0$:

$$u_j v_j = \gamma_j - \beta_j \rho. \tag{2.23}$$

Choose $z_0 \in D$ and impose the initial compatibility condition

$$\prod_{m=1}^{N} u_m(z_0)^{\beta_m} = \frac{q(z_0) + \rho'(z_0)}{2p(z_0)}. \tag{2.24}$$

Then it follows that:

$$\begin{aligned} \prod_{m=1}^{N} u_m^{\beta_m} &= \frac{q+\rho'}{2p}, \\ \prod_{m=1}^{N} v_m^{\beta_m} &= \frac{q-\rho'}{2p}, \end{aligned} \tag{2.25}$$

Proof. The hypotheses permit holomorphic logarithms of the nonvanishing factors on D. From (2.22) we may write:

$$\frac{1}{2}\sum_{j=1}^{N} \beta_j \left(\log\left[\frac{u_j}{v_j}\right]\right)' = \frac{Q}{2}\sum_{j=1}^{N}\frac{\beta_j^2}{x_j} - \sum_{j=1}^{N}\beta_j \partial_{x_j} Q + \frac{Q}{P}\sum_{j=1}^{N}\beta_j \partial_{x_j} P, \tag{2.26}$$

and from (2.23):

$$\beta_j = -\frac{{x_j}'}{\rho'}$$

hence, by making use of general calculus identities such as:

$$q' = \sum_{j=1}^{N} {x_j}' \partial_{x_j} Q$$

we may rewrite (2.26) as:

$$\begin{aligned} \left(\log\left[\prod_{j=1}^{N}\left(\frac{u_j}{v_j}\right)^{\beta_j}\right]\right)' &= -\frac{Q}{\rho'}\left(\log\left[\prod_{j=1}^{N} {x_j}^{\beta_j}\right]\right)' + \frac{2}{\rho'}\sum_{j=1}^{N} {x_j}' \partial_{x_j} Q - \frac{2Q}{\rho' P}\sum_{j=1}^{N} {x_j}' \partial_{x_j} P, \\ &= -\frac{qr'}{\rho' r} + \frac{2q'}{\rho'} - \frac{2qp'}{\rho' p}. \end{aligned} \tag{2.27}$$

However, we also observe that:

$$\begin{aligned} \left(\log\left[\frac{q+\rho'}{q-\rho'}\right]\right)' &= \frac{q'+\rho''}{q+\rho'} - \frac{q'-\rho''}{q-\rho'}, \\ &= -\frac{qr'}{\rho' r} + \frac{2q'}{\rho'} - \frac{2qp'}{\rho' p} \end{aligned} \tag{2.28}$$

hence:

$$\left(\log\left[\prod_{j=1}^{N}\left(\frac{u_j}{v_j}\right)^{\beta_j}\right]\right)' = \left(\log\left[\frac{q+\rho'}{q-\rho'}\right]\right)',$$

$$\left(\log\left[\prod_{j=1}^{N}\left(\frac{u_j}{v_j}\right)^{\beta_j}\right]\right)' + \left(\log\left[\prod_{j=1}^{N}(u_jv_j)^{\beta_j}\right]\right)' = \left(\log\left[\frac{q+\rho'}{q-\rho'}\right]\right)' + (\log[r])',$$

$$2\left(\log\left[\prod_{j=1}^{N}{u_j}^{\beta_j}\right]\right)' = \left(\log\left[r\left(\frac{q+\rho'}{q-\rho'}\right)\right]\right)',$$

$$2\left(\log\left[\prod_{j=1}^{N}{u_j}^{\beta_j}\right]\right)' = \left(\log\left[\frac{(q+\rho')^2}{4p^2}\right]\right)',$$

$$\left(\log\left[\prod_{j=1}^{N}{u_j}^{\beta_j}\right]\right)' = \left(\log\left[\frac{q+\rho'}{p}\right]\right)'. \tag{2.29}$$

Integrating and accounting for (2.24) thus gives:

$$\int_{z_0}^{z}\left(\log\left[\prod_{j=1}^{N}{u_j}^{\beta_j}\right]\right)' dz = \int_{z_0}^{z}\left(\log\left[\frac{q+\rho'}{p}\right]\right)' dz,$$

$$\prod_{j=1}^{N}{u_j}^{\beta_j} = \frac{q+\rho'}{2p}. \tag{2.30}$$

As (2.21) may be factored into:

$$-\left(\frac{\rho'+q}{2p}\right)\left(\frac{\rho'-q}{2p}\right) = r = \prod_{m=1}^{N}(u_mv_m)^{\beta_m} = \left(\prod_{m=1}^{N}u_m^{\beta_m}\right)\left(\prod_{m=1}^{N}v_m^{\beta_m}\right), \tag{2.31}$$

it follows that:

$$\prod_{j=1}^{N}{v_j}^{\beta_j} = \frac{q-\rho'}{2p}. \tag{2.32}$$

□

Initial reconstruction data. Condition (2.24) fixes one multiplicative reconstruction constant. Its right hand side is nonzero because $(q+\rho')(q-\rho') = 4p^2r \neq 0$. Since $\sum_j \beta_j \neq 0$, at least one weight is nonzero; after choosing the other nonzero initial mode values, that mode can be chosen to satisfy (2.24), with a compatible root if necessary, and $v_j(z_0) = x_j(z_0)/u_j(z_0)$. For nonconstant holomorphic solutions, the product identities extend by continuity across isolated zeros of ρ' wherever the modes and coefficients extend regularly.

Constant modal powers. Constant solutions of the squared scalar equation are not covered by the theorem: its derivation divides by ρ', and a scalar root alone need not give a compatible mode solution. Suppose $\rho(z) \equiv \rho_*$, where ρ_* is a fixed constant, and all modal products are nonzero. Then $x_j = \gamma_j - \beta_j\rho_*$ and hence p, q, r are constant. In this case the required product identities reduce to

$$U(z) = V(z) = \frac{q}{2p}, \qquad U(z) = \prod_j u_j(z)^{\beta_j}, \quad V(z) = \prod_j v_j(z)^{\beta_j}. \tag{2.33}$$

Thus the weighted products must be constant in this constant-power case, although they generally vary when ρ is nonconstant. In addition to the initial product condition and $q^2 = 4p^2 r$, compatibility requires

$$\sum_{j=1}^{N} \beta_j \left[\frac{\beta_j q}{2x_j} - \partial_{x_j} Q + \frac{q}{p} \partial_{x_j} P \right] = 0, \tag{2.34}$$

with all terms evaluated at $x_j = \gamma_j - \beta_j \rho_*$. For constant modal powers, the left hand side is U'/U. Its vanishing ensures that the initially imposed value $U = q/(2p)$ is preserved; since $UV = r$, the value of V is then preserved as well. The individual modes may still evolve, with their logarithmic derivatives cancelling in the weighted products.

3 Generating integrable systems from an autonomous differential equation

We have shown in Section 2 that we may pick Q, P and obtain from them an integrable system in which the equations of motion for u_j, v_j, ρ are separable. In some cases we may know the Q, P from which we wish to start, for example we may be working with elliptic functions and we may recognise or choose a Hamiltonian as some rearrangement of the Frobenius-Stickelberger determinant. However, we may on other occasions wish to generate a new integrable system from some known autonomous differential equation and we may not know Q, P at the outset. In this section we show how we may get Q, P quite simply from an autonomous differential equation, thereby generating an integrable system from it.

Let f be holomorphic on an open set $\mathcal{U} \subset \mathbb{C}$, and let $\rho : D \to \mathcal{U}$ be a local holomorphic solution of the autonomous equation

$$\rho' = f(\rho) \tag{3.1}$$

Following the notation of the main theorem in which $x_j = u_j v_j = \gamma_j - \beta_j \rho$ we write:

$$f(\rho) = F(x_1, \ldots, x_N). \tag{3.2}$$

The embedding of ρ into the x_j in (3.2) may manifest in two distinct ways. Firstly, it may be that F treats each x_j differently, for example if $f(\rho)$ was a polynomial in ρ, F may be a polynomial that raises each x_j to different powers but still upon substitution of the x_j for ρ produces the correct coefficients from f, this is common in physical nonlinear optics cases with elliptic function solutions for example. Alternatively, we may simply substitute the sum over $u_j v_j$ from (2.8) into f to define F, which is a simple approach for symmetric toy model exploration. In either case, from (2.21) we may then write:

$$\rho'^2 = q^2 - 4p^2 r, \tag{3.3}$$

$$f(\rho)^2 = q^2 - 4p^2 r, \tag{3.4}$$

and for consistency set:

$$F(x_1, \ldots, x_N)^2 = Q(x_1, \ldots, x_N)^2 - 4P(x_1, \ldots, x_N)^2 R(x_1, \ldots, x_N) \tag{3.5}$$

with:

$$R(x_1, \ldots, x_N) = \prod_{m=1}^{N} x_m^{\beta_m}. \tag{3.6}$$

We may solve for Q with $\varsigma = \pm 1$:

$$Q = \varsigma \sqrt{F^2 + 4P^2 R}, \tag{3.7}$$

and use this to replace Q in (2.1) yielding the Hamiltonian:

$$H\left(u_1,\dots,u_N,v_1,\dots,v_N\right) = \\ -\varsigma\sqrt{F(u_1v_1,\dots,u_Nv_N)^2+4P(u_1v_1,\dots,u_Nv_N)^2R(u_1v_1,\dots,u_Nv_N)} \\ +P(u_1v_1,\dots,u_Nv_N)\left(\prod_{m=1}^{N}u_m^{\beta_m}+\prod_{m=1}^{N}v_m^{\beta_m}\right). \quad (3.8)$$

The equations of motion can again be derived using (2.5) where one would substitute (3.7) and note:

$$\begin{aligned}\partial_{x_j}Q &= \varsigma\partial_{x_j}\sqrt{F^2+4P^2R},\\ &= \frac{\varsigma}{2\sqrt{F^2+4P^2R}}\left(2F\partial_{x_j}F+8PR\partial_{x_j}P+4P^2\partial_{x_j}R\right),\\ &= \frac{\varsigma}{\sqrt{F^2+4P^2R}}\left(F\partial_{x_j}F+4PR\partial_{x_j}P+\frac{2\beta_jP^2R}{x_j}\right).\end{aligned} \quad (3.9)$$

where, having obtained the equations of motion we again work on the zero energy level such that $H=0$ and (2.2) still holds. The same calculation as in Section 2 shows that the system again conserves the Hamiltonian and the Manley-Rowe like intermodal power laws that stem from (2.7). Furthermore, the differential equations for u_j, v_j indeed yield the target:

$$\rho'^2 = f(\rho)^2, \quad (3.10)$$

where the branch of $\rho' = \pm f(\rho)$ is fixed by the initial condition. We note that although Q is substituted in terms of F, one may still choose P subject to the local regularity conditions, so f does not generate a unique system. In particular, F must have a holomorphic extension to the modal-power domain, and F^2+4P^2R must admit the selected holomorphic square root defining Q. A nonzero radicand at the initial point is sufficient after shrinking the domain. The reconstruction formulas require P and the modal products to be nonzero; the direct theorem applies where $f(\rho)\neq 0$, with stationary cases treated separately as above.

Lemma 3.1 (Removal of a common modal-power factor)**.** *Consider the Hamiltonian in* (2.1) *on a region where* $P(x_1,\dots,x_N)\neq 0$*, and define*

$$\widehat{Q}(x_1,\dots,x_N) = \frac{Q(x_1,\dots,x_N)}{P(x_1,\dots,x_N)}. \quad (3.11)$$

Then

$$H = P\widehat{H}, \qquad \widehat{H} = \prod_{m=1}^{N}u_m^{\beta_m}+\prod_{m=1}^{N}v_m^{\beta_m}-\widehat{Q}(x_1,\dots,x_N). \quad (3.12)$$

On the level set $H=0$*, the solutions generated by* H *and* $\widehat{H}$ *describe the same curves in the variables* u_j, v_j *after a local reparameterisation of the independent variable.*

Proof. Since $P\neq 0$, the level sets $H=0$ and $\widehat{H}=0$ coincide. Hamilton's equations and the product rule give

$$\begin{aligned}\frac{du_j}{dz} &= P\frac{\partial\widehat{H}}{\partial v_j}+\widehat{H}\frac{\partial P}{\partial v_j},\\ \frac{dv_j}{dz} &= -P\frac{\partial\widehat{H}}{\partial u_j}-\widehat{H}\frac{\partial P}{\partial u_j}.\end{aligned} \quad (3.13)$$

Consequently, on $\widehat{H} = 0$,

$$\frac{du_j}{dz} = P\frac{\partial \widehat{H}}{\partial v_j}, \qquad \frac{dv_j}{dz} = -P\frac{\partial \widehat{H}}{\partial u_j}. \tag{3.14}$$

Define a new local parameter ξ along a solution by

$$\frac{d\xi}{dz} = P(x_1(z), \ldots, x_N(z)). \tag{3.15}$$

If $\widehat{u}_j(\xi) = u_j(z(\xi))$, $\widehat{v}_j(\xi) = v_j(z(\xi))$, and $\widehat{x}_j = \widehat{u}_j\widehat{v}_j$, then (3.14) becomes

$$\frac{d\widehat{u}_j}{d\xi} = \frac{\partial \widehat{H}}{\partial \widehat{v}_j}, \qquad \frac{d\widehat{v}_j}{d\xi} = -\frac{\partial \widehat{H}}{\partial \widehat{u}_j}. \tag{3.16}$$

Equivalently, the inverse change of parameter is determined by

$$z(\xi) = z_0 + \int_{\xi_0}^{\xi} \frac{d\eta}{P(\widehat{x}_1(\eta), \ldots, \widehat{x}_N(\eta))}. \tag{3.17}$$

Thus $\widehat{H} = H/P$ generates the same solution curves on the zero-energy level with the common factor P removed from the parametrisation. □

The reparameterised mode equations are explicitly

$$\begin{aligned} \frac{d\widehat{u}_j}{d\xi} &= -\widehat{u}_j\partial_{\widehat{x}_j}\widehat{Q} + \frac{\beta_j}{\widehat{v}_j}\prod_{m=1}^{N}\widehat{v}_m^{\beta_m}, \\ \frac{d\widehat{v}_j}{d\xi} &= \widehat{v}_j\partial_{\widehat{x}_j}\widehat{Q} - \frac{\beta_j}{\widehat{u}_j}\prod_{m=1}^{N}\widehat{u}_m^{\beta_m}. \end{aligned} \tag{3.18}$$

Define $\widehat{\rho}(\xi) = \rho(z(\xi))$. Then $\widehat{x}_j = \gamma_j - \beta_j\widehat{\rho}$, and the associated scalar equation is

$$\left(\frac{d\widehat{\rho}}{d\xi}\right)^2 = \widehat{Q}^{\,2} - 4\prod_{m=1}^{N}\left(\gamma_m - \beta_m\widehat{\rho}\right)^{\beta_m}. \tag{3.19}$$

Thus the factorisation is always available locally by taking $\widehat{Q} = Q/P$, but it does not in general make the original choice of P irrelevant: P may remain within the ratio Q/P and it fixes the relation between z and ξ through (3.15). The result is most useful when P and Q have an explicit common factor which cancels from $\widehat{Q}$, in which case that factor affects only the parametrisation on the chosen energy level. The modular λ system in Section 9 provides such an example, since the common factor involving the complete elliptic integral can be transferred into the change of parameter. For broader discussion of phase-space-dependent multiples of Hamiltonian vector fields as time reparameterisations, see [24]. The same argument applies on any fixed energy level after subtracting the energy from H, as is implemented here by an additive change to Q.

3.1 Recognising equivalent Hamiltonian forms

A Hamiltonian that depends nonlinearly on the mixing terms may also admit a representation within the family considered here on a chosen level set. To see this, write $x = (x_1, \ldots, x_N)$, $U = \prod_j u_j^{\beta_j}$, $V = \prod_j v_j^{\beta_j}$, and consider

$$\mathcal{H}_S = P(x)S(U + V) - Q(x). \tag{3.20}$$

Work locally where $P \neq 0$ and S admits a differentiable local inverse with nonvanishing derivative; in the complex setting, take a regular holomorphic inverse branch. On the corresponding branch of $\mathcal{H}_S = 0$, applying S^{-1} gives

$$U + V = h(x), \qquad h(x) = S^{-1}\left(\frac{Q(x)}{P(x)}\right). \tag{3.21}$$

Thus the same local level set is described by $K = U + V - h(x) = 0$, a Hamiltonian of the original form with coupling function 1. Equality of the level sets alone does not identify their parametrised dynamics. Indeed, differentiating $S(h) = Q/P$ shows that, on this level,

$$d\mathcal{H}_S = A(x)\, dK, \qquad A(x) = P(x)S'(h(x)), \tag{3.22}$$

where d denotes the differential on phase space. Their Hamiltonian vector fields therefore satisfy $X_{\mathcal{H}_S} = AX_K$, so the solution curves agree after the local change of parameter $d\xi/dz = A(x(z))$, as in Lemma 3.1. Alternatively, define

$$\widetilde{H} = A(x)K = \widetilde{P}(x)(U + V) - \widetilde{Q}(x), \qquad \widetilde{P} = A, \quad \widetilde{Q} = Ah. \tag{3.23}$$

Since $d\widetilde{H} = A\, dK$ on $K = 0$, this Hamiltonian reproduces the vector field of $\mathcal{H}_S$ on the selected level in the original variable z. Consequently, a locally invertible function of $U + V$ does not enlarge the class of level-set dynamics represented by arbitrary P, Q, although this equivalence may not be apparent from the initial Hamiltonian expression. The equivalence is local to the chosen inverse branch and level set; the Hamiltonians need not generate the same dynamics away from that level.

4 Formulation in terms of total complex modal phase

Throughout this section, all expressions are considered locally with consistent branch choices and where the displayed denominators are nonzero. We know that via (2.11) we may write the modal power in terms of $\rho(z)$. Therefore, if we define a complex modal phase function $\phi_j : \mathbb{C} \to \mathbb{C}$, we can write the modes as:

$$\begin{aligned} u_j(z) &= \sqrt{\gamma_j - \beta_j \rho(z)}\, e^{\phi_j(z)}, \\ v_j(z) &= \sqrt{\gamma_j - \beta_j \rho(z)}\, e^{-\phi_j(z)}. \end{aligned} \tag{4.1}$$

We then define a total complex modal phase $\Phi : \mathbb{C} \to \mathbb{C}$, weighted by degeneracy integers $\beta_m \in \mathbb{Z}$ $(m = 1, \ldots, N)$, as:

$$\Phi(z) = \sum_{j=1}^{N} \beta_j \phi_j(z) \tag{4.2}$$

From (2.25) we get:

$$\begin{aligned} \prod_{j=1}^{N} \left(\frac{u_j}{v_j}\right)^{\beta_j} &= \frac{q + \rho'}{q - \rho'}, \\ e^{2\Phi} &= \frac{q + f(\rho)}{q - f(\rho)} \end{aligned} \tag{4.3}$$

and thus $\Phi(z)$ can be expressed as a function of ρ and thus as a function of modal powers. We also get the useful relations:

$$\begin{aligned} q &= \frac{\rho'}{\tanh \Phi}, \\ p &= \frac{\rho'}{2\sqrt{r} \sinh \Phi}, \end{aligned} \tag{4.4}$$

or equivalently:

$$\cosh\Phi = \frac{q}{2\sqrt{r}\,p},$$
$$\sinh\Phi = \frac{\rho'}{2\sqrt{r}\,p}. \tag{4.5}$$

We also have by the definition of ϕ_j and Φ the relations:

$$\cosh\Phi = \frac{\left(\prod_{m=1}^{N} u_m^{\beta_m} + \prod_{m=1}^{N} v_m^{\beta_m}\right)}{2\sqrt{\prod_{j=1}^{N}(u_j v_j)^{\beta_j}}},$$
$$\sinh\Phi = \frac{\left(\prod_{m=1}^{N} u_m^{\beta_m} - \prod_{m=1}^{N} v_m^{\beta_m}\right)}{2\sqrt{\prod_{j=1}^{N}(u_j v_j)^{\beta_j}}}. \tag{4.6}$$

This viewpoint can be useful when constructing a Hamiltonian system from predefined solutions. In particular, it enables us to write a Hamiltonian system in which the solutions are related to the predefined solutions by quadrature, in a way that can be thought of as a gauge-like transform. Let us suppose we start from predefined solutions u_j, v_j which we will refer to hereafter as the prescribed solutions. Let $\theta_j : \mathbb{C}^N \to \mathbb{C}$ be a function we choose when constructing the prescribed solutions to encode how each of the mode phases depends on all of the mode powers. The existence of such θ_j might appear presumptive and restrictive, but on any interval where ρ is locally invertible it is formally possible to write any z dependent phase term as a function of ρ via $z = \rho^{-1}(\rho(z))$, and then replace ρ with a function of modal powers. Let $\hat{\Phi} : \mathbb{C}^N \to \mathbb{C}$ be the degeneracy weighted sum of mode phases which encodes how the total modal phase depends on individual mode powers. Similarly to how q, p relate to Q, P we thus set:

$$\phi_j(z) = \theta_j(x_1(z), \ldots, x_N(z)),$$
$$\Phi(z) = \hat{\Phi}\left(x_1(z), \ldots, x_N(z)\right) = \sum_{j=1}^{N} \beta_j \theta_j(x_1(z), \ldots, x_N(z)). \tag{4.7}$$

Suppose then we use the functions $\hat{\Phi}$, θ_j, and F to build a Hamiltonian from (2.1), (3.2), and (4.4), such that:

$$H\left(\bar{u}_1, \ldots, \bar{u}_N, \bar{v}_1, \ldots, \bar{v}_N\right) = \left(\prod_{m=1}^{N} \bar{u}_m^{\beta_m} + \prod_{m=1}^{N} \bar{v}_m^{\beta_m}\right) \frac{F(\bar{u}_1\bar{v}_1, \ldots, \bar{u}_N\bar{v}_N)}{2\sqrt{\prod_{j=1}^{N}(\bar{u}_j\bar{v}_j)^{\beta_j}}\,\sinh\hat{\Phi}\left(\bar{u}_1\bar{v}_1, \ldots, \bar{u}_N\bar{v}_N\right)}$$
$$- \frac{F(\bar{u}_1\bar{v}_1, \ldots, \bar{u}_N\bar{v}_N)}{\tanh\hat{\Phi}\left(\bar{u}_1\bar{v}_1, \ldots, \bar{u}_N\bar{v}_N\right)}, \tag{4.8}$$

where we denote the modes in this Hamiltonian as $\bar{u}_j, \bar{v}_j$ and we will enquire whether $u_j = \bar{u}_j, v_j = \bar{v}_j$. It is important to stress that (4.3) and thus (4.4) and (4.5) hold on the level set $H = 0$. For clarity, the way to build and use the system in (4.8) from prescribed solutions would be as follows:

- Decide the autonomous differential equation that gives $\rho' = f(\rho) = F(x_1, \ldots, x_N)$. Together with the constants β_j, γ_j and an initial condition, this determines the modal powers x_j.
- Decide how the mode phases ϕ_j are to depend on $x_1, \ldots, x_N$ and thus on ρ, then sum them weighted by degeneracy to get $\hat{\Phi}(x_1, \ldots, x_N)$ - this now already determines both the solutions and thus the equations of motion in the prescribed system but this may not in general be the Hamiltonian system we are constructing.

- Use the constructed $\hat{\Phi}$ function as $\hat{\Phi}\left(\bar{u}_1\bar{v}_1, \ldots, \bar{u}_N\bar{v}_N\right)$ in (4.8) but crucially, without using (4.3), (4.4) and (4.5) to further simplify the Hamiltonian as otherwise it would trivially become zero prior to differentiating.
- Apply Hamilton's equations to (4.8) to obtain the general equations of motion for $\bar{u}_j, \bar{v}_j$ and only after doing so does one work on the level set $H = 0$ and use (4.3), (4.4) and (4.5) to simplify the equations of motion.

Writing $\bar{x}_j = \bar{u}_j\bar{v}_j$, the equations corresponding to (2.14) are

$$\begin{aligned}
\frac{d\log\bar{u}_j}{dz} &= -F\partial_{\bar{x}_j}\hat{\Phi} - \frac{\beta_j F}{2\bar{x}_j}, \\
\frac{d\log\bar{v}_j}{dz} &= F\partial_{\bar{x}_j}\hat{\Phi} - \frac{\beta_j F}{2\bar{x}_j},
\end{aligned} \tag{4.9}$$

where F and $\hat{\Phi}$ are evaluated at $(\bar{x}_1, \ldots, \bar{x}_N)$. These equations imply

$$\bar{x}_j' = -\beta_j F. \tag{4.10}$$

Choose a reference point z_0 at which the Hamiltonian modes and prescribed modes agree:

$$\bar{u}_j(z_0) = u_j(z_0), \qquad \bar{v}_j(z_0) = v_j(z_0). \tag{4.11}$$

It follows that $\bar{x}_j(z_0) = x_j(z_0)$. Since the prescribed and Hamiltonian modal powers satisfy the same differential equations with the same initial values, they agree on the corresponding local solution branch:

$$\bar{x}_j = x_j. \tag{4.12}$$

We henceforth make this identification. The partial derivatives of $\hat{\Phi}$ determine the individual Hamiltonian modal-phase evolution:

$$\frac{1}{2}\frac{d\log\left(\frac{\bar{u}_j}{\bar{v}_j}\right)}{dz} = \bar{\phi}_j' = -F\partial_{x_j}\hat{\Phi} = -F\sum_{k=1}^{N}\beta_k\partial_{x_j}\theta_k. \tag{4.13}$$

However, by definition the prescribed individual phase velocity is:

$$\phi_j' = -F\sum_{k=1}^{N}\beta_k\partial_{x_k}\theta_j. \tag{4.14}$$

Thus, if we define

$$\xi_j'(z) = \bar{\phi}_j' - \phi_j' = -F\sum_{k=1}^{N}\beta_k\left(\partial_{x_j}\theta_k - \partial_{x_k}\theta_j\right), \tag{4.15}$$

then the constructed Hamiltonian yields the prescribed phase velocities precisely when

$$\sum_{k=1}^{N}\beta_k\left(\partial_{x_j}\theta_k - \partial_{x_k}\theta_j\right) = 0, \qquad j = 1, \ldots, N. \tag{4.16}$$

Otherwise, the solutions of the constructed Hamiltonian system are related to the prescribed solutions by

$$\begin{aligned}
\bar{u}_j &= u_j e^{\xi_j}, \\
\bar{v}_j &= v_j e^{-\xi_j}.
\end{aligned} \tag{4.17}$$

The common initial values at z_0 give $\xi_j(z_0) = 0$. Let η_j express this phase correction as a function of ρ, such that

$$\xi_j(z) = \eta_j(\rho(z)). \tag{4.18}$$

Letting $\rho_0 = \rho(z_0)$, we have $\eta_j(\rho_0) = 0$, and using $F = \rho'$ the phase correction is given by

$$\begin{aligned} \frac{d\eta_j}{d\rho} &= -\sum_{k=1}^{N} \beta_k \left(\partial_{x_j}\theta_k - \partial_{x_k}\theta_j\right), \\ \eta_j(\rho) &= -\int_{\rho_0}^{\rho} \sum_{k=1}^{N} \beta_k \left(\partial_{x_j}\theta_k - \partial_{x_k}\theta_j\right) ds. \end{aligned} \tag{4.19}$$

In the integrand, all partial derivatives are evaluated at

$$(x_1, \ldots, x_N) = (\gamma_1 - \beta_1 s, \ldots, \gamma_N - \beta_N s). \tag{4.20}$$

Thus the solutions of the Hamiltonian system are known by quadrature and can be written explicitly in terms of the prescribed modes. Furthermore, antisymmetry under interchange of j and k gives

$$\sum_{j=1}^{N} \beta_j \xi_j' = -F \sum_{j=1}^{N} \sum_{k=1}^{N} \beta_j \beta_k \left(\partial_{x_j}\theta_k - \partial_{x_k}\theta_j\right) = 0. \tag{4.21}$$

Since $\xi_j(z_0) = 0$, it follows that

$$\sum_{j=1}^{N} \beta_j \xi_j = 0 \tag{4.22}$$

for all z. The transformation in (4.17) therefore preserves each modal power and both product terms in the Hamiltonian:

$$\begin{aligned} \bar{u}_j \bar{v}_j &= u_j v_j, \\ \prod_{j=1}^{N} \bar{u}_j^{\beta_j} &= \prod_{j=1}^{N} u_j^{\beta_j}, \\ \prod_{j=1}^{N} \bar{v}_j^{\beta_j} &= \prod_{j=1}^{N} v_j^{\beta_j}. \end{aligned} \tag{4.23}$$

Hence the Hamiltonian in (4.8) is exactly invariant along the paired solutions, and the transformation in (4.17) can be regarded as gauge-like.

An example of building the Hamiltonian in this way is given in Section 9.

5 Nonlinear optics, classical mechanics, and elliptic functions

Nonlinear optics operating in the quasi-continuous-wave regime where time derivatives are negligible in comparison to other effects provided much of the inspiration for this research. These systems often have solutions in terms of doubly periodic and quasi-doubly periodic elliptic functions. We will present a general system that encompasses many of these systems as special cases and then list important examples in a table. We defer the proof of their solutions to existing literature.

For constants $a_j, a_{j,k}, c \in \mathbb{C}$, let us consider the $N \leq 4$ mode system with:

$$\begin{aligned} Q\left(x_1, \ldots, x_N\right) &= a_0 + \sum_{j=1}^{N} a_j x_j + \frac{1}{2} \sum_{j=1,k=1}^{N} a_{j,k} x_j x_k, \\ P\left(x_1, \ldots, x_N\right) &= c \end{aligned} \tag{5.1}$$

such that by (2.1) the Hamiltonian is:

$$H\left(u_1,\ldots,u_N,v_1,\ldots,v_N\right)=c\left(\prod_{m=1}^{N}u_m^{\beta_m}+\prod_{m=1}^{N}v_m^{\beta_m}\right)-a_0-\sum_{j=1}^{N}a_ju_jv_j-\frac{1}{2}\sum_{j=1,k=1}^{N}a_{j,k}u_jv_ju_kv_k \tag{5.2}$$

with $a_{j,k}=a_{k,j}$, and Hamilton's equations give:

$$\begin{aligned}\frac{\partial}{\partial z}u_j&=\partial_{v_j}H=-\left(a_j+\sum_{k=1}^{N}a_{j,k}u_kv_k\right)u_j+\frac{c\beta_j}{v_j}\prod_{k=1}^{N}v_k^{\beta_k},\\ \frac{\partial}{\partial z}v_j&=-\partial_{u_j}H=\left(a_j+\sum_{k=1}^{N}a_{j,k}u_kv_k\right)v_j-\frac{c\beta_j}{u_j}\prod_{k=1}^{N}u_k^{\beta_k}.\end{aligned} \tag{5.3}$$

As before, after deriving the equations of motion we restrict to the zero-energy level $H=0$; the constant a_0 offsets this energy level within Q.

By (2.11) and (2.16) we have:

$$\rho'^2=\left(a_0+\sum_{j=1}^{N}a_j\left(\gamma_j-\beta_j\rho\right)+\sum_{j,k=1}^{N}\frac{a_{j,k}}{2}\left(\gamma_j-\beta_j\rho\right)\left(\gamma_k-\beta_k\rho\right)\right)^2-4c^2\prod_{j=1}^{N}\left(\gamma_j-\beta_j\rho\right)^{\beta_j} \tag{5.4}$$

and thus, provided we limit ourselves to values of $a_{j,k}$, N and β_j that ensure the right hand side of (5.4) is a cubic or quartic in ρ, then ρ is generically an elliptic function of z and u_j,v_j can be expressed in terms of Weierstrass σ, ζ, and $\wp$ (but may be multivalued). The system in (5.3) contains as special cases many well-known systems from nonlinear optics, elliptic function theory, and classical mechanics, which we summarise in the table below where, unless further specified, Q,P take the most general forms given in (5.1).

N	β	**Integrable System**
4	$\beta_j = 1$	Four-wave mixing in cubic nonlinear material [8, 20, 2, 3, 4, 5].
3	$\beta_1 = \beta_2 = 1$, $\beta_3 = 2$	Degenerate four-wave mixing in cubic nonlinear material [9, 5].
3	$\beta_j = 1$	• Three-wave mixing in quadratic nonlinear material where $a_{j,k} = 0$ but a_j may be non-zero [7, 1, 19]. • Projection from u_j, v_j to the Lagrange top [10], where $a_{j,k} = 0$ but $a_j \neq 0$ and special initial conditions and parameter values apply • Jacobi elliptic functions system for $\mathrm{sn}, \mathrm{cn}, \mathrm{dn}$ [25], the Euler top [10], and a projection from u_j, v_j to the Duffing oscillator [11], where $a_j = a_{j,k} = 0$ and special initial conditions apply such that $v_j \propto u_j$
2	$\beta_1 = 2$, $\beta_2 = 2$	Orthogonal linearly polarised modes in cubic nonlinear material [7, 5].
2	$\beta_j = 1$	Asymmetric and symmetric coherent couplers in cubic nonlinear material, both Hermitian and Parity Time symmetric [9, 6].
2	$\beta_1 = 2$, $\beta_2 = 1$	Two-wave mixing in quadratic nonlinear material where $a_{j,k} = 0$ but a_j may be non-zero [7, 1].
1	$\beta_1 = 4$	A system that can be thought of as the lemniscate exponential in the sense that $u_j(z) = \mathrm{cl}(z) + i\,\mathrm{sl}(z)$ and $v_j(z) = \mathrm{cl}(z) - i\,\mathrm{sl}(z)$, i.e., a version of Euler's formula for the exponential using lemniscate sine and cosine $(\mathrm{sl}, \mathrm{cl})$ [25], where special initial conditions and parameter values apply.
1	$\beta_1 = 3$	Dixon elliptic functions [26] where $a_{j,k} = 0$ but $a_j \neq 0$ and special initial conditions and parameter values apply.

5.1 Higher-degree scalar curves and hyperelliptic solutions

The elliptic restriction in the preceding examples is a restriction on the degree of the scalar polynomial, rather than on the mode count alone. Retaining $P = c \neq 0$, allow arbitrary N, positive integer weights β_j, and a polynomial Q of total degree d. On the conserved power line, put

$$q(\rho) = Q(\gamma_1 - \beta_1\rho, \ldots, \gamma_N - \beta_N\rho), \qquad B = \sum_{j=1}^{N} \beta_j. \tag{5.5}$$

The same reduction gives

$$\rho'^2 = F(\rho), \qquad F(\rho) = q(\rho)^2 - 4c^2 \prod_{j=1}^{N} (\gamma_j - \beta_j\rho)^{\beta_j}. \tag{5.6}$$

Here $\deg q \leq d$, and $\deg F \leq \max(2d, B)$; its actual degree must be checked after restriction and cancellation of leading terms. If F is squarefree of degree $m \geq 5$, the smooth compactification of $y^2 = F(\rho)$ has genus $g = \lfloor (m-1)/2 \rfloor \geq 2$. Thus quintic and sextic scalar polynomials give genus two, while septic and octic polynomials give genus three. Repeated roots instead require normalization of a singular curve and can lower the genus.

The scalar motion is locally determined by the hyperelliptic Abelian integral

$$z - z_0 = \int_{\rho(z_0)}^{\rho(z)} \frac{dt}{\sqrt{F(t)}}, \tag{5.7}$$

with the sheet and continuation fixed by the initial data. Higher genus does not, in general, give a globally single-valued meromorphic inverse in z. In the companion manuscript [21], submitted to arXiv with identifier and posting date pending, we construct local solutions for this polynomial Hamiltonian class using hyperelliptic sigma functions and Abelian integrals, for smooth odd-degree curves subject to the stated degree and regularity conditions. The scalar sigma representation lies on the one-point Abel image: only the Abel coordinate associated with $d\rho/y$ is linear in z, while the remaining coordinates are constrained by that image. The individual modes require further reconstruction from their powers. In that setting, Ônishi's hyperelliptic determinant identities [23] encode the paired-power relations and complete weighted mode products entering the Hamiltonian, extending our genus-one connection with the Frobenius–Stickelberger determinant [8, 9].

5.1.1 The N-extended Euler system

A useful special case is the N-extended Euler system studied by Ferrer, Crespo, and Molero [27]. Set $\beta_j = 1$, take Q constant, and impose $v_j = \kappa_j u_j$ with nonzero constants κ_j. For nontrivial motion, compatibility of the two canonical equations requires $\prod_j \kappa_j = -1$. On $H = 0$ this also requires $Q = 0$, and the restricted equations become

$$u_j' = \alpha_j \prod_{k \neq j} u_k, \qquad \alpha_j = -\frac{c}{\kappa_j}. \tag{5.8}$$

Their quadratic first integrals are the restrictions of the intermodal power laws: $\kappa_j u_j^2 - \kappa_k u_k^2 = \gamma_j - \gamma_k$. For $N = 3$, choose $\alpha_1 = 1$, $\alpha_2 = -1$, and $\alpha_3 = -k^2$. Initial data $(u_1, u_2, u_3) = (0, 1, 1)$ then give $(\mathrm{sn}, \mathrm{cn}, \mathrm{dn})$ of modulus k. For general N, the scalar polynomial in (5.6) reduces to $-4c^2 \prod_j (\gamma_j - \rho)$, of degree N. With distinct γ_j, its genus is $\lfloor (N-1)/2 \rfloor$: $N = 4$ is still elliptic, whereas $N = 5$ gives genus two. Ferrer, Crespo, and Molero examine the $N = 4, 5$ cases and reparameterisations that generalise the Jacobi amplitude. Their constant-Q, proportional-mode systems therefore sit within the broader polynomial-Q canonical family considered here.

5.1.2 A one-mode Fermat family and incomplete beta inversion

A complementary special case has one canonical pair $u(z), v(z)$ and a single interaction weight $\beta_1 = n > 1$. For integers n, m, with $n > 1$, take

$$H(u, v) = [1 - 4(uv)^n]^m \left(u^n + v^n - 1\right), \qquad P(x) = Q(x) = (1 - 4x^n)^m. \tag{5.9}$$

We work on the Fermat component $u^n + v^n = 1$, locally where the prefactor is finite and nonzero. A nonzero constant Fermat level can be normalized to 1 by rescaling the modes and independent variable. For $m = 0$, $P = Q = 1$; the remaining integers introduce a common prefactor, polynomial for $m \geq 0$ and rational for $m < 0$.

The canonical scalar coordinate is $\rho(z) = -u(z)v(z)/n$. Define the fixed polynomial $D(\rho) = 1 - 4(-n\rho)^n$. On $H = 0$,

$$\rho' = D(\rho)^m (u^n - v^n), \qquad \rho'^2 = D(\rho)^{2m+1}, \qquad u^n = \frac{1 + \rho'/D(\rho)^m}{2}, \quad v^n = \frac{1 - \rho'/D(\rho)^m}{2}. \tag{5.10}$$

Compatible roots recover the modes with $uv = -n\rho$. To exhibit the inversion, choose a real local branch with $u, v > 0$ and $u^n > v^n$, and put $s(z) = 4[-n\rho(z)]^n$. Then $\rho = -s^{1/n}/(n4^{1/n})$ and separation gives

$$B_{s(z)}\left(\frac{1}{n}, \frac{1}{2} - m\right) - B_{s(z_0)}\left(\frac{1}{n}, \frac{1}{2} - m\right) = -n^2 4^{1/n}(z - z_0), \tag{5.11}$$

where $B_s(a, b) = \int_0^s t^{a-1}(1-t)^{b-1}\,dt$ is the unregularized incomplete beta function. Its local inverse determines $s(z)$, hence ρ and the mode powers. For $m \geq 1$, the second beta parameter is negative, but the integral remains finite for $0 < s < 1$.

For $m \geq 1$, the scalar polynomial has repeated roots, so its high degree does not represent a generic higher-genus curve. Removing its square factor by $Y = \rho'/D(\rho)^m$ gives $Y^2 = D(\rho)$, whose smooth compactification has genus $\lfloor (n-1)/2 \rfloor$. The same normalization applies to the rational scalar relation for $m < 0$. Thus this is a degenerate algebraic scalar family with an incomplete-beta parametrisation; its normalized scalar curve is distinct from the full Fermat curve, whose genus is $(n-1)(n-2)/2$.

6 Log-augmented polynomial systems

Throughout this section, dynamical variables are functions of the independent variable z, and primes denote differentiation with respect to z. We display this dependence when introducing variables and suppress it subsequently for readability. Parameters and fixed values of first integrals are constant along each trajectory.

We introduce the category of log-augmented polynomial systems to describe polynomial mode equations whose scalar generating equations retain logarithmic terms. The scalar equations belong to a broader class of rational functions augmented by powers of a logarithm, which also includes systems with rational mode equations. A useful form is

$$\rho'(z)^2 = R_1(\rho(z)) + R_2(\rho(z))\big[\log R_3(\rho(z))\big]^m, \qquad m \in \mathbb{Z}, \tag{6.1}$$

where R_1, R_2, R_3 are fixed rational functions of their scalar argument and m is a constant integer. More generally, several logarithmic powers may occur together. We work locally on consistent logarithmic branches, away from poles and other singularities. Such equations can arise in the Hamiltonian framework considered here when logarithmic terms $[\log \mathcal{R}(x_1, \ldots, x_N)]^M$ are combined with rational functions of the modal powers $x_j = u_j v_j$ and the weighted products $U = \prod_j u_j^{\beta_j}$, $V = \prod_j v_j^{\beta_j}$, for choices compatible with the scalar reduction. The Hamiltonian exponent M need not equal m: squaring or eliminating the mode products can generate mixed logarithmic powers.

These systems are interesting because their equations of motion can be polynomial or rational even though their scalar reductions contain logarithms. Differentiation of logarithmic Hamiltonian terms produces rational expressions, while the zero-energy relation can eliminate remaining logarithms from the mode equations. The logarithmic dependence nevertheless persists in the conserved relation and reappears in the scalar equation for ρ. Thus, unlike the systems in the preceding section, polynomial mode equations here need not imply an elliptic or hyperelliptic scalar reduction: the logarithmic terms generally prevent the reduced relation from defining an algebraic curve. The examples below illustrate how this structure can remain hidden in otherwise familiar-looking polynomial systems.

We first present an $m = 1$ reduction of a polynomial system considered by Sergyeyev [14] and a separate rational-coefficient equation with an explicit Lambert W solution. We then consider four systems involving squared logarithmic expressions, possibly with shifted or rescaled arguments and lower logarithmic powers after expansion.

6.1 First-power logarithmic terms: $m = 1$

6.1.1 A polynomial system considered by Sergyeyev

In Example 2 of [14], Sergyeyev applies coupling constant metamorphosis to a system attributed to Goriely and obtains the polynomial system for $X(z), Y(z), Z(z)$, with constant parameter b:

$$X' = -2X^2 + 2Z, \qquad Y' = -3XY, \qquad Z' = -2XZ + 2X^3 + 6bX. \tag{6.2}$$

Its first integrals have constant values I, J:

$$I = \frac{b + X^2 - Z}{Y^2}, \qquad J = \frac{X^2}{2} + Z + b\log(Y^2). \tag{6.3}$$

Sergyeyev uses these to reduce the system to a scalar equation containing a first power of a logarithm. To relate it to the present mode variables, choose a local square root $Z(z) = \widehat{Z}(z)^2$ and set

$$u(z) = X(z) - \widehat{Z}(z), \qquad v(z) = X(z) + \widehat{Z}(z), \qquad x(z) = u(z)v(z) = X(z)^2 - Z(z). \tag{6.4}$$

For $I \neq 0$, the integral J becomes

$$J = \frac{3}{8}(u^2 + v^2) - \frac{x}{4} + b\log\left(\frac{x+b}{I}\right). \tag{6.5}$$

The first integral defines a fixed function g of the modal power, whose value $g(u(z)v(z))$ varies along the trajectory:

$$g(uv) = \sqrt{\frac{8}{3}J - \frac{4}{3}uv - \frac{8b}{3}\log\left(\frac{uv+b}{I}\right)}. \tag{6.6}$$

Choose its local branch so that $g(uv) = v - u$ on the conserved surface, away from its zeros and the logarithmic singularities. A single Hamiltonian of the form (2.1), with $N = 1$ and $\beta_1 = 2$, is then

$$H(u, v) = \frac{uv+b}{g(uv)}\left[-\frac{3}{2}(u^2 + v^2) + uv + 4J - 4b\log\left(\frac{uv+b}{I}\right)\right]. \tag{6.7}$$

On $H = 0$, its canonical equations reproduce the transformed (X, Z) subsystem in the original independent variable z. The remaining variable is recovered from $Y^2 = (uv + b)/I$ with a consistent branch. We give this Hamiltonian representation for completeness of the examples; we make no particular claim that it offers an advantage over Sergyeyev's original formulation.

In the original variable z, direct differentiation gives $(uv)' = -3(uv+b)(u+v)$. Our one-mode convention with $\beta_1 = 2$ fixes $\gamma_1 = 0$ and hence

$$\rho(z) = -\frac{u(z)v(z)}{2}. \tag{6.8}$$

Using the first integral to eliminate $(u + v)^2$ then gives

$$\rho'^2 = 6(b - 2\rho)^2\left[J - 2\rho - b\log\left(\frac{b-2\rho}{I}\right)\right]. \tag{6.9}$$

This is an $m = 1$ member of (6.1), with polynomial R_1, R_2 and $R_3(\rho) = (b - 2\rho)/I$, retaining the canonical modal-power coordinate ρ and the original independent variable z.

6.1.2 An explicit Lambert W example

Independently of the preceding system, let a be a nonzero constant and consider the rational-coefficient equation for $\rho(z)$

$$\rho'^2 = \frac{4\rho^2}{(a\rho-1)^2}\,(a\rho - \log\rho)\,, \qquad a \neq 0. \tag{6.10}$$

A local solution family is

$$\rho(z) = -\frac{1}{a}\,W_k\left(-ae^{-(z-z_0)^2}\right), \tag{6.11}$$

where z_0 is a constant of integration, k is a fixed branch index, and W_k is the corresponding branch of the Lambert function [28]. Indeed, its defining identity yields

$$a\rho - \log\rho = (z-z_0)^2, \qquad \rho' = \frac{2(z-z_0)\rho}{a\rho-1}, \tag{6.12}$$

which verify (6.10). The logarithmic branch is chosen to satisfy the first identity, and the formula is used locally away from $a\rho = 1$ and other branch singularities. This supplies an explicitly solvable $m = 1$ equation of the stated rational-coefficient form.

6.2 Squared logarithmic terms: $m = 2$

The elliptic examples in the previous section are generated by scalar differential equations which can be reduced to the Weierstrass equation for $s(z)$, with constant invariants g_2, g_3

$$s'^2 = 4s^3 - g_2 s - g_3. \tag{6.13}$$

We now consider systems generated by a logarithmic deformation of this equation. A convenient factorised form of the deformation for $\rho(z)$, with constant coefficients c_0, c_1, c_2, c_3, c_5, is

$$\rho'^2 = c_0(\rho-c_1)(\rho-c_2)(\rho-c_3) + c_5\rho^2\log^2\rho. \tag{6.14}$$

Under the linear change of variable

$$\rho(z) = \frac{4s(z)}{c_0} + \frac{c_1+c_2+c_3}{3} = \frac{4(s(z)+c_6)}{c_0}, \tag{6.15}$$

with $c_0 \neq 0$ and the constant $c_6 = (c_0/12)(c_1+c_2+c_3)$, equation (6.14) takes the Weierstrass-like form

$$s'^2 = 4s^3 - g_2 s - g_3 + c_5(s+c_6)^2\log^2\left(\frac{4(s+c_6)}{c_0}\right). \tag{6.16}$$

We refer to (6.16) as the logarithmically deformed Weierstrass equation. Setting $c_5 = 0$ recovers the ordinary Weierstrass equation. It is also useful to note that the reciprocal change $\xi(z) = 1/\rho(z)$ gives

$$\xi'^2 = -c_0\xi\prod_{j=1}^{3}(c_j\xi - 1) + c_5\xi^2\log^2\xi. \tag{6.17}$$

We will show that the Lotka–Volterra model, a parity-time symmetric dimer, and two further mode systems are related through special or degenerate cases of these equations. Appendix B further clarifies their relation to elliptic systems by expressing the logarithmically deformed equation as a Weierstrass function of a reparameterised variable, where the change of variable is itself determined by a non-elliptic integral.

6.2.1 The Lotka–Volterra model

Let $\widehat{u}(z), \widehat{v}(z)$ satisfy the two-species Lotka–Volterra equations. The parameters $\alpha, \beta, \kappa, \delta$ are constant, and we use the sign convention

$$\begin{aligned} \widehat{u}' &= \alpha\widehat{u} - \delta\widehat{u}\widehat{v}, \\ \widehat{v}' &= -\beta\widehat{v} - \kappa\widehat{u}\widehat{v}. \end{aligned} \tag{6.18}$$

For $\kappa, \delta \neq 0$, introduce the rescaled variables

$$\widehat{u}(z) = \frac{u(z)}{\kappa}, \qquad \widehat{v}(z) = -\frac{v(z)}{\delta}, \tag{6.19}$$

which satisfy

$$\begin{aligned} u' &= \alpha u + uv, \\ v' &= -\beta v - uv. \end{aligned} \tag{6.20}$$

The rescaled system conserves

$$\mathcal{K}(u, v) = \alpha \log(-v) + \beta \log u + u + v. \tag{6.21}$$

Fixing a value K of this conserved quantity, define the canonical Hamiltonian

$$H_{\mathrm{LV}} = uv\left[\alpha \log(-v) + \beta \log u + u + v - K\right]. \tag{6.22}$$

Hamilton's equations give

$$\begin{aligned} u' &= \partial_v H_{\mathrm{LV}} = \frac{H_{\mathrm{LV}}}{v} + \alpha u + uv, \\ v' &= -\partial_u H_{\mathrm{LV}} = -\frac{H_{\mathrm{LV}}}{u} - \beta v - uv. \end{aligned} \tag{6.23}$$

Thus, locally where $uv \neq 0$, the zero-energy level $H_{\mathrm{LV}} = 0$ reproduces (6.20).

In the symmetric case $\beta = \alpha$, the Hamiltonian becomes

$$H_{\mathrm{LV}} = (u + v)uv - (K - \alpha \log(-uv))\, uv. \tag{6.24}$$

This is the $N = 1$, $\beta_1 = 1$ case of (2.1). Writing $x(z) = u(z)v(z)$, the fixed power-dependent functions are

$$P(x) = x, \qquad Q(x) = (K - \alpha \log(-x))\, x. \tag{6.25}$$

Our convention $\sum_{j=1}^{N} \gamma_j = 0$ fixes the additive freedom in ρ. For $N = 1$ this necessarily gives $\gamma_1 = 0$, so that $\rho = -uv$, and equation (2.16) becomes

$$\rho'^2 = 4\rho^3 + \rho^2 \left(K - \alpha \log \rho\right)^2. \tag{6.26}$$

For $\alpha \neq 0$, this is a special case of (6.16): one may take $g_2 = g_3 = c_6 = 0$, $c_5 = \alpha^2$, and choose the scale inside the logarithm so that $c_0 = 4e^{K/\alpha}$.

6.2.2 A parity-time symmetric dimer

Ramezani et al. [29] study a nonlinear optical dimer with balanced gain and loss and cubic Kerr nonlinearity. Its polynomial mode equations conceal the inverse-trigonometric conservation

law that produces the logarithmic scalar reduction below. We use its complexified form for $\psi_1(z), \psi_2(z), \widetilde{\psi}_1(z), \widetilde{\psi}_2(z)$, with constant parameters χ, κ:

$$\begin{aligned}
\psi_1' &= \mathrm{i}\chi\psi_1^2\widetilde{\psi}_1 + \kappa\psi_1 + \mathrm{i}\psi_2, \\
\psi_2' &= \mathrm{i}\chi\psi_2^2\widetilde{\psi}_2 - \kappa\psi_2 + \mathrm{i}\psi_1, \\
\widetilde{\psi}_1' &= -\mathrm{i}\chi\psi_1\widetilde{\psi}_1^2 + \kappa\widetilde{\psi}_1 - \mathrm{i}\widetilde{\psi}_2, \\
\widetilde{\psi}_2' &= -\mathrm{i}\chi\psi_2\widetilde{\psi}_2^2 - \kappa\widetilde{\psi}_2 - \mathrm{i}\widetilde{\psi}_1.
\end{aligned} \tag{6.27}$$

The physical dimer is recovered by $\widetilde{\psi}_j = \psi_j^*$; χ is the Kerr coefficient and κ the balanced gain/loss coefficient, with the linear coupling normalized to unity. Introduce the Stokes variables $S_0(z), S_1(z), S_2(z)$:

$$\begin{aligned}
S_0 &= \psi_1\widetilde{\psi}_1 + \psi_2\widetilde{\psi}_2, \\
S_1 &= \psi_1\widetilde{\psi}_2 + \psi_2\widetilde{\psi}_1, \\
S_2 &= \mathrm{i}\left(\psi_1\widetilde{\psi}_2 - \psi_2\widetilde{\psi}_1\right).
\end{aligned} \tag{6.28}$$

In the notation used here, the first integrals of [29], together with a locally equivalent arctangent representation, are

$$\begin{aligned}
C^2 &= (\chi S_1 - 2)^2 + (\chi S_2)^2, \\
J &= S_0 + \frac{2\kappa}{\chi}\arcsin\left(\frac{\chi S_1 - 2}{C}\right) \\
&= S_0 - \frac{2\kappa}{\chi}\arctan\left(\frac{\chi S_1 - 2}{\chi S_2}\right).
\end{aligned} \tag{6.29}$$

For real variables, the principal inverse-sine expression generally requires $C\chi S_2 < 0$ for conservation; extension beyond this sign domain requires continuation of its branch. The cited paper specifies initial data and signed values of C, but does not state this branch restriction explicitly. The arctangent form likewise requires a continued angle. For the complexified system, a logarithmic representation makes the conservation law more direct:

$$\begin{aligned}
C^2 &= 4\left(\chi\psi_1\widetilde{\psi}_2 - 1\right)\left(\chi\psi_2\widetilde{\psi}_1 - 1\right), \\
J &= S_0 - \frac{2\mathrm{i}\kappa}{\chi}\log\left(\frac{\mathrm{i}C}{2(\chi\psi_1\widetilde{\psi}_2 - 1)}\right).
\end{aligned} \tag{6.30}$$

With compatible branches, (6.30) recovers the local inverse-trigonometric forms in (6.29). Direct differentiation gives $J' = 0$ for the complexified equations wherever the logarithmic expression is defined, without the real sign restriction above; the logarithm must be continued along the trajectory rather than reset to its principal branch. In its logarithmic form, J has the same structure as the Lotka–Volterra conservation law: a sum of modal powers together with a logarithm.

We now relate (6.27) to a canonical Hamiltonian in the coordinates of the present theory. For one choice of the square-root branch, introduce canonical modes $u_j(z), v_j(z)$, $j = 1, 2$, and consider the $N = 2$, $\beta_1 = \beta_2 = 1$ Hamiltonian

$$\begin{aligned}
H_{\mathrm{PT}} = \frac{\sqrt{2}}{4}(u_1u_2 + v_1v_2)\sqrt{C^2 - 4u_1v_1 - 4u_2v_2 + 4} \\
+ \frac{\mathrm{i}}{8}(C^2 - 4u_1v_1 - 4u_2v_2 + 4)\left[J\chi + 2\mathrm{i}\kappa\log\left(\frac{-4\mathrm{i}C}{C^2 - 4u_1v_1 - 4u_2v_2 + 4}\right)\right].
\end{aligned} \tag{6.31}$$

The associated intermodal power law may be written

$$C^2 = 4u_1v_1 - 4u_2v_2 + 4. \tag{6.32}$$

On $H_{\mathrm{PT}} = 0$, define a phase function $\varphi(z)$ by

$$\varphi' = -\frac{3\mathrm{i}}{4}\left[J\chi + 2\mathrm{i}\kappa \log\left(\frac{-4\mathrm{i}C}{C^2 - 4u_1v_1 - 4u_2v_2 + 4}\right)\right], \tag{6.33}$$

and make the coordinate transformation

$$\begin{aligned} \psi_1 &= \frac{v_2 e^{-\varphi}}{\sqrt{\chi}}, & \psi_2 &= \frac{2\mathrm{i}u_1 e^{-\varphi}}{\sqrt{\chi}\sqrt{C^2 - 4u_1v_1}}, \\ \widetilde{\psi}_1 &= \frac{2\mathrm{i}v_1 e^{\varphi}}{\sqrt{\chi}\sqrt{C^2 - 4u_1v_1}}, & \widetilde{\psi}_2 &= \frac{u_2 e^{\varphi}}{\sqrt{\chi}}. \end{aligned} \tag{6.34}$$

Direct substitution of Hamilton's equations for (6.31) into (6.34) gives (6.27).

As usual, set $x_j = u_j v_j$ and

$$\rho = -\frac{x_1 + x_2}{2}, \qquad x_j = \gamma_j - \rho, \tag{6.35}$$

where the intermodal power law fixes

$$\gamma_1 = \frac{C^2}{8} - \frac{1}{2}, \qquad \gamma_2 = -\gamma_1. \tag{6.36}$$

The scalar equation generated by (6.31) is

$$\begin{aligned} \rho'^2 = -\frac{(C^2 + 8\rho + 4)^2}{64}\left[J\chi + 2\mathrm{i}\kappa \log\left(\frac{-4\mathrm{i}C}{C^2 + 8\rho + 4}\right)\right]^2 \\ + \frac{(C^2 - 8\rho - 4)(C^2 + 8\rho - 4)(C^2 + 8\rho + 4)}{128}. \end{aligned} \tag{6.37}$$

Define the function $w(z)$ and the constant A by

$$w(z) = C^2 + 8\rho(z) + 4, \qquad A = -4\mathrm{i}C \exp\left(\frac{J\chi}{2\mathrm{i}\kappa}\right), \tag{6.38}$$

this becomes

$$w'^2 = -\frac{1}{2}w(w - 8)(w - 2C^2) + 4\kappa^2 w^2 \log^2\left(\frac{w}{A}\right), \tag{6.39}$$

which is again of the form (6.14), up to a constant rescaling of its argument.

The connection between (6.26), (6.37), and the logarithmically deformed Weierstrass equation suggests further mode systems of the same type.

6.2.3 A three-mode logarithmic system

Take $N = 3$, $\beta_j = 1$, and modes $u_j(z), v_j(z)$ with modal powers $x_j(z) = u_j(z)v_j(z)$. Let a_0, a_1, a_2, a_3 be constants, and define

$$\rho = -\frac{u_1v_1 + u_2v_2 + u_3v_3}{3}. \tag{6.40}$$

Define the canonical Hamiltonian

$$\begin{aligned} H_3 = \left[-\frac{a_0}{3}(u_1v_1 + u_2v_2 + u_3v_3) - a_1\right] \\ \times \log\left[a_3 - \frac{a_2}{3}(u_1v_1 + u_2v_2 + u_3v_3)\right] + u_1u_2u_3 + v_1v_2v_3. \end{aligned} \tag{6.41}$$

With constants γ_j fixed by the initial powers, this has the two independent intermodal power laws

$$\begin{aligned} u_1 v_1 - u_2 v_2 &= \gamma_1 - \gamma_2, \\ u_1 v_1 - u_3 v_3 &= \gamma_1 - \gamma_3, \end{aligned} \tag{6.42}$$

and on $H_3 = 0$ generates

$$\rho'^2 = (a_0\rho - a_1)^2 \log^2(a_2\rho + a_3) - 4(\gamma_1 - \rho)(\gamma_2 - \rho)(\gamma_3 - \rho). \tag{6.43}$$

The general equation (6.43) allows different linear functions of ρ outside and inside the logarithm. The restriction $a_1 = a_3 = 0$ is directly of the form (6.14).

In this restricted case, introduce new functions $\widetilde{u}_j(z), \widetilde{v}_j(z)$ and $S(z)$, locally where $S(z) \neq 0$, by

$$u_j(z) = S(z)\widetilde{u}_j(z), \qquad v_j(z) = S(z)\widetilde{v}_j(z), \qquad S = \widetilde{u}_1\widetilde{v}_1 + \widetilde{u}_2\widetilde{v}_2 + \widetilde{u}_3\widetilde{v}_3. \tag{6.44}$$

Then $\rho = -S^3/3$, and the transformed equations take the polynomial form

$$\begin{aligned} \widetilde{u}_1' &= (-\widetilde{u}_1\widetilde{v}_1 + \widetilde{u}_2\widetilde{v}_2 + \widetilde{u}_3\widetilde{v}_3)\,\widetilde{v}_2\widetilde{v}_3 - \frac{a_0}{3}\widetilde{u}_1, \\ \widetilde{u}_2' &= (\widetilde{u}_1\widetilde{v}_1 - \widetilde{u}_2\widetilde{v}_2 + \widetilde{u}_3\widetilde{v}_3)\,\widetilde{v}_1\widetilde{v}_3 - \frac{a_0}{3}\widetilde{u}_2, \\ \widetilde{u}_3' &= (\widetilde{u}_1\widetilde{v}_1 + \widetilde{u}_2\widetilde{v}_2 - \widetilde{u}_3\widetilde{v}_3)\,\widetilde{v}_1\widetilde{v}_2 - \frac{a_0}{3}\widetilde{u}_3, \\ \widetilde{v}_1' &= -(-\widetilde{u}_1\widetilde{v}_1 + \widetilde{u}_2\widetilde{v}_2 + \widetilde{u}_3\widetilde{v}_3)\,\widetilde{u}_2\widetilde{u}_3 + \frac{a_0}{3}\widetilde{v}_1, \\ \widetilde{v}_2' &= -(\widetilde{u}_1\widetilde{v}_1 - \widetilde{u}_2\widetilde{v}_2 + \widetilde{u}_3\widetilde{v}_3)\,\widetilde{u}_1\widetilde{u}_3 + \frac{a_0}{3}\widetilde{v}_2, \\ \widetilde{v}_3' &= -(\widetilde{u}_1\widetilde{v}_1 + \widetilde{u}_2\widetilde{v}_2 - \widetilde{u}_3\widetilde{v}_3)\,\widetilde{u}_1\widetilde{u}_2 + \frac{a_0}{3}\widetilde{v}_3. \end{aligned} \tag{6.45}$$

Each nonlinear term is a product of four mode amplitudes, making the resemblance to a five-wave-mixing system visible directly from the equations. The intermodal power laws become

$$\begin{aligned} (\widetilde{u}_1\widetilde{v}_1 - \widetilde{u}_2\widetilde{v}_2)\,S^2 &= \gamma_1 - \gamma_2, \\ (\widetilde{u}_1\widetilde{v}_1 - \widetilde{u}_3\widetilde{v}_3)\,S^2 &= \gamma_1 - \gamma_3. \end{aligned} \tag{6.46}$$

6.2.4 A two-mode logarithmic system

For a second example, take $N = 2$, $\beta_1 = \beta_2 = 1$, modes $u_1(z), u_2(z), v_1(z), v_2(z)$, and constant parameters a_0, a_1, a_2. Define the canonical Hamiltonian

$$H_2 = (a_1 u_1 v_1 + a_2 u_2 v_2)\left[\log\left(a_0(a_1 u_1 v_1 + a_2 u_2 v_2)\right) - u_1 u_2 - v_1 v_2\right]. \tag{6.47}$$

On $H_2 = 0$, Hamilton's equations simplify to the polynomial system

$$\begin{aligned} u_1' &= a_1 u_1 - (a_1 u_1 v_1 + a_2 u_2 v_2) v_2, \\ u_2' &= a_2 u_2 - (a_1 u_1 v_1 + a_2 u_2 v_2) v_1, \\ v_1' &= (a_1 u_1 v_1 + a_2 u_2 v_2) u_2 - a_1 v_1, \\ v_2' &= (a_1 u_1 v_1 + a_2 u_2 v_2) u_1 - a_2 v_2. \end{aligned} \tag{6.48}$$

The nonlinear terms are cubic in the mode amplitudes, as in four-wave mixing in a cubic medium. Writing $x_1 = u_1 v_1$ and $x_2 = u_2 v_2$, the modal powers satisfy $x_1' - x_2' = 0$. Taking

$$\rho = -\frac{x_1 + x_2}{2}, \tag{6.49}$$

we may therefore write

$$x_j = \gamma_j - \rho, \qquad \gamma_1 + \gamma_2 = 0. \tag{6.50}$$

For compactness in the scalar equation only, define the fixed affine function R by

$$R(\rho) = a_1(\gamma_1 - \rho) + a_2(\gamma_2 - \rho). \tag{6.51}$$

Equation (2.16) gives

$$\rho'^2 = R(\rho)^2 \left[\log^2(a_0 R(\rho)) - 4(\gamma_1 - \rho)(\gamma_2 - \rho)\right]. \tag{6.52}$$

Finally, on defining

$$w(z) = \frac{1}{a_0 R(\rho(z))}, \tag{6.53}$$

equation (6.52) becomes

$$w'^2 = \frac{4}{a_0^2}\left(2a_0a_1\gamma_1 w - 1\right)\left(2a_0a_2\gamma_1 w + 1\right) + (a_1 + a_2)^2 w^2 \log^2 w. \tag{6.54}$$

This is a degenerate case of the factorised form (6.14), in which one root is taken to infinity with a compensating rescaling of the cubic coefficient.

These examples show how logarithmic scalar generating equations connect known systems and generate further mode equations. The scalar equation, the canonical coordinates, and the choice of independent variable must all be specified when comparing their solutions.

7 Rational generating equations and generalised Lambert–W-type inverse laws

Rational choices of the generating equation form a simple class for which the inverse-function viewpoint can be made explicit. We first record the correspondence between these equations and a broad family of generally multivalued inverse laws.

Theorem 7.1 (Rational generating equations)**.** *Let*

$$\rho' = f(\rho) = \frac{A(\rho)}{B(\rho)}, \tag{7.1}$$

where A and B are coprime polynomials and f is not identically zero. Every nonconstant solution is locally the inverse of a relation of the form

$$e^{\mathcal{P}(\rho)+\mathcal{R}(\rho)} \prod_{\ell=1}^{L} (\rho - a_\ell)^{\nu_\ell} = e^z, \tag{7.2}$$

where $\mathcal{P}$ is a polynomial whose constant term is fixed by the initial condition, and $\mathcal{R}$ is a rational function consisting of a finite sum of principal parts. Conversely, every local inverse defined by a relation of the form (7.2) *satisfies an autonomous differential equation in which $f(\rho)$ is rational.*

Proof. On a region where $f(\rho)$ is finite and nonzero, interchange the dependent and independent variables to obtain

$$\frac{dz}{d\rho} = \frac{B(\rho)}{A(\rho)}. \tag{7.3}$$

Over $\mathbb{C}$, its partial-fraction expansion may be written

$$\frac{B(\rho)}{A(\rho)} = p(\rho) + \sum_{\ell=1}^{L} \sum_{k=1}^{n_\ell} \frac{c_{\ell k}}{(\rho - a_\ell)^k}, \tag{7.4}$$

where p is a polynomial and the positive integer n_ℓ is the order of the pole at a_ℓ. Termwise integration gives

$$z = \mathcal{P}(\rho) + \sum_{\ell=1}^{L} c_{\ell 1} \log(\rho - a_\ell) - \sum_{\ell=1}^{L} \sum_{k=2}^{n_\ell} \frac{c_{\ell k}}{(k-1)(\rho - a_\ell)^{k-1}}, \tag{7.5}$$

where $\mathcal{P}' = p$ and the constant of integration has been included in the constant term of $\mathcal{P}$. Exponentiating (7.5) gives (7.2), with

$$\nu_\ell = c_{\ell 1}, \qquad \mathcal{R}(\rho) = -\sum_{\ell=1}^{L} \sum_{k=2}^{n_\ell} \frac{c_{\ell k}}{(k-1)(\rho - a_\ell)^{k-1}}. \tag{7.6}$$

For the converse, take a local logarithm of (7.2) and differentiate to find

$$\rho' = \left[\mathcal{P}'(\rho) + \mathcal{R}'(\rho) + \sum_{\ell=1}^{L} \frac{\nu_\ell}{\rho - a_\ell} \right]^{-1}, \tag{7.7}$$

which is rational in ρ. □

We refer to the generally multivalued local inverses in (7.2) as extended Lambert–W-type inverse functions. The classical Lambert function is recovered by taking $\mathcal{P}(\rho) = \rho$, $\mathcal{R} = 0$, $L = 1$, $a_1 = 0$, and $\nu_1 = 1$, so that (7.2) reduces to $\rho e^\rho = e^z$ [28]; an arbitrary multiplicative factor on the right is equivalent to adding a constant to $\mathcal{P}$. The generalised Lambert function of Mező and Baricz is also contained in (7.2): it corresponds to a linear $\mathcal{P}$, $\mathcal{R} = 0$, and integer powers of the linear factors [30]. Castle subsequently allowed arbitrary complex powers of the linear factors and developed Taylor series for the resulting inverse functions [31]. For this subfamily, a Taylor series $w(Z) = \sum_{n=0}^{\infty} b_n Z^n$ for the inverse function gives the corresponding solution of the rational equation as

$$\rho(z) = w(e^z) = \sum_{n=0}^{\infty} b_n e^{nz}, \tag{7.8}$$

where the series is used within its domain of convergence. The further inclusion of a general polynomial $\mathcal{P}$ and rational principal parts $\mathcal{R}$ in the exponential is what permits the partial-fraction expansion of an arbitrary rational equation to be represented by (7.2). This family is not intended to include every inverse function that has been described as a generalisation of Lambert W; Section 8 considers a different extension based on inverse spherical Bessel functions. As elsewhere in this paper, the logarithms, powers, and inverse functions are understood locally with branches fixed by the initial condition. Zeros of f also give constant solutions of (7.1); these are separate from the nonconstant locally invertible solutions described by the theorem.

A first consequence is that rational functions of $\rho(z)$ have elementary antiderivatives along a solution of (7.1). Indeed, if G is rational, then

$$\int G(\rho(z))\, dz = \int \frac{G(\rho)}{f(\rho)}\, d\rho, \tag{7.9}$$

and the integrand on the right is rational. Over $\mathbb{C}$ the result is therefore a sum of a rational function of ρ and logarithms of linear functions of ρ. This is particularly useful when solving the integrable systems generated in this way: once $\rho(z)$ is known, any rational functions of ρ that occur in the remaining separated mode equations can be integrated explicitly in elementary form.

There is also a direct consequence for the coupled systems constructed above: rational choices of F and P generate coupled first-order equations that are rational in the modes. Suppose that $F(x_1, \ldots, x_N)$ in (3.2) and the freely chosen function $P(x_1, \ldots, x_N)$ are rational. Since

the β_j are integers, $R = \prod_m x_m^{\beta_m}$ and the modal products $\prod_m u_m^{\beta_m}$ and $\prod_m v_m^{\beta_m}$ are rational functions of the modes. Although Q in (3.7) need not itself be rational away from the chosen energy level, on $H = 0$ we have

$$Q = P\left(\prod_{m=1}^{N} u_m^{\beta_m} + \prod_{m=1}^{N} v_m^{\beta_m}\right). \tag{7.10}$$

Using (7.10) to remove the square root from the equations obtained through (3.7), we obtain on any regular part of $H = 0$ a system of rational ordinary differential equations for u_j, v_j. This implication is only in one direction: a coupled system that is rational or polynomial in the modes need not have a rational generating function $f(\rho)$ and therefore need not be solved by the inverse functions in (7.2). The elliptic systems in Section 5 provide immediate examples: their mode equations may be polynomial even though $f(\rho)^2$, rather than $f(\rho)$ itself, is polynomial, and their solutions are generally elliptic functions.

Finally, any such finite rational system may be reduced locally to a polynomial system by a further change of parameter. Collecting all denominators, write the restricted equations as

$$\frac{dy_a}{dz} = \frac{A_a(y)}{D(y)}, \qquad y = (u_1, \ldots, u_N, v_1, \ldots, v_N), \qquad a = 1, \ldots, 2N, \tag{7.11}$$

where the A_a and D are polynomials. On a region where $D(y) \neq 0$, define

$$\frac{d\xi}{dz} = \frac{1}{D(y(z))}. \tag{7.12}$$

It then follows that

$$\frac{dy_a}{d\xi} = A_a(y), \tag{7.13}$$

so the rational and polynomial systems trace the same local solution curves with different parametrisations. This is the same Hamiltonian multiplication and time-reparameterisation mechanism as in Lemma 3.1, applied in the opposite direction. Multiplying H by the denominator D gives a Hamiltonian for the reparameterised polynomial equations on $H = 0$, since the additional terms produced by the product rule are proportional to H and vanish there. In particular, using a common modal-power denominator D_0 for the rational functions F and P gives a corresponding Kummer-type Hamiltonian with the polynomial functions D_0F and D_0P. An additional factor depending separately on the modes, i.e. not just mode powers, may be needed to clear every denominator in the coupled equations; the resulting Hamiltonian remains canonical on the chosen level set but need not retain the Kummer-type form away from it.

8 A system generated by an inverse spherical Bessel function

We now give a deliberately constructed example whose purpose is to demonstrate the freedom of the inverse procedure developed in Sections 2–4. Rather than beginning with a familiar Hamiltonian system, we prescribe a locally invertible special function from which we obtain the equation for ρ, the mode solutions, and finally a non-polynomial Hamiltonian that generates them.

Let j_1 denote the spherical Bessel function of the first kind of order one,

$$j_1(s) = \frac{\sin s}{s^2} - \frac{\cos s}{s}. \tag{8.1}$$

The real inverse of j_1 is multivalued, with locally single-valued branches on intervals between successive zeros of j_1' [32]. Although inverse spherical Bessel functions have themselves been described as generalisations of Lambert W, they are different from the inverse laws in Section 7.

Writing the sine and cosine in j_1 in exponential form introduces both e^{is} and e^{-is}, rather than the single exponential-product structure in (7.2); correspondingly, the function $1/j_1'(\rho)$ below is not rational in ρ. Choose c such that $j_1'(c) \neq 0$ and select the local inverse branch containing c by defining

$$j_1(\rho(z)) = j_1(c) + z, \qquad \rho(0) = c. \tag{8.2}$$

Differentiating (8.2) gives the autonomous equation

$$\begin{aligned} \rho' = f(\rho) &= \frac{1}{j_1'(\rho)}, \\ &= \frac{\rho^3}{\rho^2 \sin\rho + 2\rho\cos\rho - 2\sin\rho}. \end{aligned} \tag{8.3}$$

The solution is local: it can be continued along the chosen branch only while $j_1'(\rho) \neq 0$.

We construct a two-mode system by taking $N = 2$ and $\beta_1 = \beta_2 = 1$. Writing $x_j = u_j v_j$, the intermodal power laws give

$$x_j = \gamma_j - \rho, \qquad \gamma_1 + \gamma_2 = 0, \qquad \rho = -\frac{x_1 + x_2}{2}. \tag{8.4}$$

For constants $a_j, b_j \in \mathbb{C}$, choose the modal phases

$$\phi_j(x_j) = a_j \operatorname{Si}(b_j x_j), \qquad \operatorname{Si}(s) = \int_0^s \frac{\sin t}{t}\, dt, \tag{8.5}$$

and hence the total phase

$$\widehat{\Phi}(x_1, x_2) = a_1 \operatorname{Si}(b_1 x_1) + a_2 \operatorname{Si}(b_2 x_2). \tag{8.6}$$

This choice has the useful derivative

$$\partial_{x_j} \phi_j = a_j b_j j_0(b_j x_j), \tag{8.7}$$

where $j_0(s) = \sin(s)/s$. The prescribed modes are therefore

$$\begin{aligned} u_j(z) &= \alpha_j \sqrt{\gamma_j - \rho(z)}\, \exp\left[a_j \operatorname{Si}\left(b_j(\gamma_j - \rho(z))\right)\right], \\ v_j(z) &= \frac{1}{\alpha_j} \sqrt{\gamma_j - \rho(z)}\, \exp\left[-a_j \operatorname{Si}\left(b_j(\gamma_j - \rho(z))\right)\right], \end{aligned} \tag{8.8}$$

where $\alpha_j \neq 0$ are constants satisfying $\alpha_1 \alpha_2 = 1$, as required for the selected $H = 0$ branch of the Hamiltonian below, and all square roots and inverse functions are taken on consistent local branches.

To construct the Hamiltonian, define

$$\begin{aligned} F(x_1, x_2) = f\left(-\frac{x_1 + x_2}{2}\right) &= \frac{1}{j_1'\left(-(x_1 + x_2)/2\right)}, \\ &= \frac{1}{j_1'\left((x_1 + x_2)/2\right)}, \end{aligned} \tag{8.9}$$

where the final equality follows because j_1 is odd and j_1' is even. Applying the construction in (4.4) gives

$$P(x_1, x_2) = \frac{F(x_1, x_2)}{2\sqrt{x_1 x_2}\sinh \widehat{\Phi}(x_1, x_2)}, \qquad Q(x_1, x_2) = \frac{F(x_1, x_2)}{\tanh \widehat{\Phi}(x_1, x_2)}. \tag{8.10}$$

Thus a canonical Hamiltonian generating the prescribed power and phase behaviour is

$$H = \frac{F(x_1,x_2)(u_1u_2+v_1v_2)}{2\sqrt{x_1x_2}\sinh\widehat{\Phi}(x_1,x_2)} - \frac{F(x_1,x_2)}{\tanh\widehat{\Phi}(x_1,x_2)},$$
$$x_1 = u_1v_1, \qquad x_2 = u_2v_2. \quad (8.11)$$

Both P and Q are non-polynomial functions of the modal powers.

After applying Hamilton's equations to (8.11) and then working on the level set $H = 0$, the resulting coupled system can be written directly in terms of the modes as

$$\begin{aligned}
\frac{du_j}{dz} &= -\frac{a_jb_jj_0\left(b_ju_jv_j\right)}{j_1'\left((u_1v_1+u_2v_2)/2\right)}u_j - \frac{1}{2v_jj_1'\left((u_1v_1+u_2v_2)/2\right)},\\
\frac{dv_j}{dz} &= \frac{a_jb_jj_0\left(b_ju_jv_j\right)}{j_1'\left((u_1v_1+u_2v_2)/2\right)}v_j - \frac{1}{2u_jj_1'\left((u_1v_1+u_2v_2)/2\right)}, \qquad j=1,2. \quad (8.12)
\end{aligned}$$

Here j_1' denotes differentiation with respect to the argument of j_1. The first term in each equation is proportional to the corresponding mode and contains $j_0(b_ju_jv_j)$, a function of that mode's own modal power. It may therefore be interpreted as a generalised self-phase-modulation-like term whose overall strength is modulated by the common total-power factor in the denominator. The second terms determine the modal-power evolution, and direct substitution into (8.12) gives

$$(u_jv_j)' = -\frac{1}{j_1'\left((u_1v_1+u_2v_2)/2\right)} = -\rho', \quad (8.13)$$

in agreement with (8.4) and (8.3).

Figure 1 compares the modes in (8.8) with a direct numerical integration of the four coupled equations in (8.12). The numerical points lie on the prescribed solutions up to the end of the plotted inverse branch. The increasingly rapid variation near the right endpoint occurs as ρ approaches the first critical point of j_1 on that branch, where $j_1'(\rho) = 0$. This example is intentionally removed from the usual polynomial and elliptic settings: it shows directly how the choice of a locally invertible function can generate a solvable canonical Hamiltonian system involving a different class of special functions.

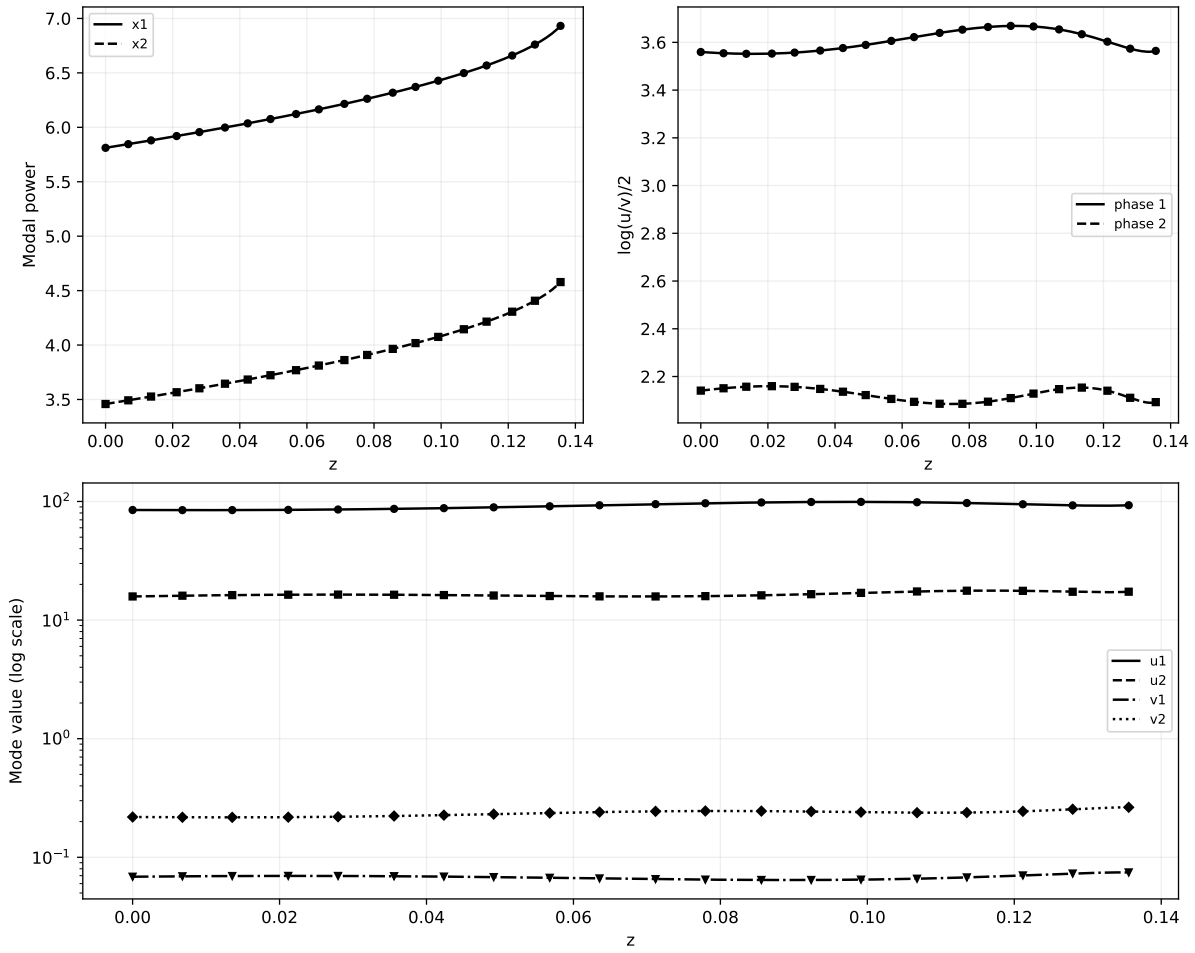


Figure 1: Comparison of the prescribed inverse-spherical-Bessel solutions (lines) with direct numerical integration of the coupled Hamiltonian system (markers). The parameters are $a_1 = 2.3$, $a_2 = 1.35$, $b_1 = 6.4$, and $b_2 = 9.7$, with $x_1(0) = 5.8116$, $x_2(0) = 3.4584$, and $\alpha_1 = \alpha_2 = 1$ in (8.8). Thus $u_j(0) = \sqrt{x_j(0)}\exp[a_j\,\mathrm{Si}(b_j x_j(0))]$ and $v_j(0) = x_j(0)/u_j(0)$, giving $\rho(0) = -4.635$ on the selected branch of j_1^{-1}. Numerical integration uses SciPy `solve_ivp` with DOP853, relative tolerance 10^{-11}, absolute tolerance 10^{-13}, and maximum step 0.05. The comparison stops at 98% of the interval to the next critical point of j_1; the mode panel uses a logarithmic vertical scale.

9 A system generated by the modular λ function

We now give a second deliberately constructed example, this time generated by the modular lambda function. Its first-order differential equation is

$$\frac{d\lambda}{d\tau} = \frac{4i}{\pi}\lambda(1-\lambda)K(\lambda)^2, \tag{9.1}$$

where $K(m)$ denotes the complete elliptic integral of the first kind with parameter m. The familiar theta-constant representation is

$$\lambda(\tau) = \frac{\vartheta_2(\tau)^4}{\vartheta_3(\tau)^4}, \qquad \lambda(\tau) - 1 = -\frac{\vartheta_4(\tau)^4}{\vartheta_3(\tau)^4}. \tag{9.2}$$

Here and below the theta constants have nome $q = e^{i\pi\tau}$. This example is useful because the generating equation is classical and explicitly solvable, while the resulting functions P and Q are non-polynomial functions of the modal powers.

Take $N = 2$ and $\beta_1 = \beta_2 = 2$, and write

$$x_1 = u_1 v_1, \qquad x_2 = u_2 v_2. \tag{9.3}$$

For parameters $a_0, a_2 \in \mathbb{C}$, choose

$$\begin{aligned} P(x_1, x_2) &= \frac{i}{\pi} \sinh(a_2) K\left(\frac{1 + x_1 + x_2}{2}\right)^2, \\ Q(x_1, x_2) &= \frac{2i}{\pi} K\left(\frac{1 + x_1 + x_2}{2}\right)^2 \left[\cosh(a_2) x_1 x_2 + a_0(x_1 - x_2 - 1)\right]. \end{aligned} \tag{9.4}$$

The Hamiltonian (2.1) is therefore

$$\begin{aligned} H = \frac{i}{\pi} K\left(\frac{1 + u_1 v_1 + u_2 v_2}{2}\right)^2 \Big[&\sinh(a_2)(u_1^2 u_2^2 + v_1^2 v_2^2) - 2\cosh(a_2) u_1 u_2 v_1 v_2 \\ &- 2a_0(u_1 v_1 - u_2 v_2 - 1)\Big]. \end{aligned} \tag{9.5}$$

Apart from the common elliptic-integral factor, the interaction has the quartic form familiar from degenerate four-wave mixing, while the a_0 term is linear in the modal powers.

Hamilton's equations for (9.5) initially also contain derivatives of K. On the level set $H = 0$, the bracketed expression in (9.5) vanishes, so all terms obtained by differentiating the common elliptic-integral factor vanish. The equations then reduce to the comparatively simple system

$$\begin{aligned} \frac{du_1}{d\tau} &= \frac{2i}{\pi} K\left(\frac{1 + u_1 v_1 + u_2 v_2}{2}\right)^2 \left[\sinh(a_2) v_1 v_2^2 - \cosh(a_2) u_1 u_2 v_2 - a_0 u_1\right], \\ \frac{du_2}{d\tau} &= \frac{2i}{\pi} K\left(\frac{1 + u_1 v_1 + u_2 v_2}{2}\right)^2 \left[\sinh(a_2) v_2 v_1^2 - \cosh(a_2) u_2 u_1 v_1 + a_0 u_2\right], \\ \frac{dv_1}{d\tau} &= \frac{2i}{\pi} K\left(\frac{1 + u_1 v_1 + u_2 v_2}{2}\right)^2 \left[-\sinh(a_2) u_1 u_2^2 + \cosh(a_2) v_1 u_2 v_2 + a_0 v_1\right], \\ \frac{dv_2}{d\tau} &= \frac{2i}{\pi} K\left(\frac{1 + u_1 v_1 + u_2 v_2}{2}\right)^2 \left[-\sinh(a_2) u_2 u_1^2 + \cosh(a_2) v_2 u_1 v_1 - a_0 v_2\right]. \end{aligned} \tag{9.6}$$

Thus the non-polynomial dependence appears only through a common function of the two modal powers, while the remaining nonlinearities are cubic in the modes.

The two modal powers obey

$$\begin{aligned} \frac{d(u_1 v_1)}{d\tau} &= \frac{d(u_2 v_2)}{d\tau} \\ &= -\frac{2i}{\pi} \sinh(a_2) K\left(\frac{1 + u_1 v_1 + u_2 v_2}{2}\right)^2 (u_1 u_2 - v_1 v_2)(u_1 u_2 + v_1 v_2), \end{aligned} \tag{9.7}$$

and hence $x_1 - x_2$ is conserved. In the notation of the general theory,

$$x_j = \gamma_j - 2\rho, \qquad \gamma_1 + \gamma_2 = 0, \qquad \rho = -\frac{x_1 + x_2}{4}. \tag{9.8}$$

We select the invariant manifold

$$\gamma_1 = \frac{1}{2}, \qquad \gamma_2 = -\frac{1}{2}, \qquad x_1 - x_2 = 1. \tag{9.9}$$

On the component of $H = 0$ for which $u_1 u_2/(v_1 v_2) = \coth(a_2/2)$, the scalar equation is

$$\rho' = \frac{2i}{\pi}\left(-2\rho - \frac{1}{2}\right)\left(-2\rho + \frac{1}{2}\right) K\left(\frac{1}{2} - 2\rho\right)^2. \tag{9.10}$$

The change of variable

$$\rho(\tau) = \frac{1}{4} - \frac{1}{2}\lambda(\tau - \tau_0) \tag{9.11}$$

turns (9.10) exactly into (9.1). Consequently,

$$u_1 v_1 = \lambda(\tau - \tau_0), \qquad u_2 v_2 = \lambda(\tau - \tau_0) - 1. \tag{9.12}$$

Locally, τ_0 is fixed by the initial modal power through the inverse relation

$$\tau_0 = -i\frac{K(1 - u_1(0)v_1(0))}{K(u_1(0)v_1(0))}, \tag{9.13}$$

with the modular and elliptic-integral branches chosen consistently.

The individual modes are also explicit. Integrating the phase equations gives

$$\begin{aligned} \frac{u_1}{v_1} &= \alpha_1 \left(\frac{\lambda(\tau - \tau_0) - 1}{\lambda(\tau - \tau_0)}\right)^{a_0}, \\ \frac{u_2}{v_2} &= \alpha_2 \left(\frac{\lambda(\tau - \tau_0) - 1}{\lambda(\tau - \tau_0)}\right)^{-a_0}, \end{aligned} \tag{9.14}$$

where α_1 and α_2 are non-zero constants. Combining (9.14) with (9.12) yields the following lambda-function and theta-constant forms. To keep the display readable, every occurrence of λ and ϑ_j in this equation is evaluated at $\tau - \tau_0$:

$$\begin{aligned} u_1^2 &= \alpha_1 \lambda \left(\frac{\lambda - 1}{\lambda}\right)^{a_0} = \alpha_1 \frac{\vartheta_2^4}{\vartheta_3^4} \left(-\frac{\vartheta_4^4}{\vartheta_2^4}\right)^{a_0}, \\ u_2^2 &= \alpha_2 (\lambda - 1) \left(\frac{\lambda - 1}{\lambda}\right)^{-a_0} = -\alpha_2 \frac{\vartheta_4^4}{\vartheta_3^4} \left(-\frac{\vartheta_4^4}{\vartheta_2^4}\right)^{-a_0}, \\ v_1^2 &= \frac{\lambda}{\alpha_1} \left(\frac{\lambda - 1}{\lambda}\right)^{-a_0} = \frac{\vartheta_2^4}{\alpha_1 \vartheta_3^4} \left(-\frac{\vartheta_4^4}{\vartheta_2^4}\right)^{-a_0}, \\ v_2^2 &= \frac{\lambda - 1}{\alpha_2} \left(\frac{\lambda - 1}{\lambda}\right)^{a_0} = -\frac{\vartheta_4^4}{\alpha_2 \vartheta_3^4} \left(-\frac{\vartheta_4^4}{\vartheta_2^4}\right)^{a_0}. \end{aligned} \tag{9.15}$$

These expressions are understood locally with consistent logarithmic and square-root branches. The remaining condition $H = 0$ relates a_2 to the integration constants:

$$\tanh(a_2) = \frac{2\alpha_1 \alpha_2}{\alpha_1^2 \alpha_2^2 + 1}. \tag{9.16}$$

Equivalently, away from limiting cases, the two possible branches are $\alpha_1 \alpha_2 = \tanh(a_2/2)$ and $\alpha_1 \alpha_2 = \coth(a_2/2)$.

The second equality in each line of (9.15) follows from (9.2). In particular, when a_0 is a half-integer, the square roots can be chosen so that the modes are ratios of integer powers of theta constants and are therefore meromorphic functions of τ. For example, for $a_0 = 1/2$ one consistent choice is

$$\begin{aligned} u_1 &= e^{i\pi/4} \sqrt{\alpha_1} \frac{\vartheta_2 \vartheta_4}{\vartheta_3^2}, & u_2 &= e^{i\pi/4} \sqrt{\alpha_2} \frac{\vartheta_2 \vartheta_4}{\vartheta_3^2}, \\ v_1 &= e^{-i\pi/4} \frac{1}{\sqrt{\alpha_1}} \frac{\vartheta_2^3}{\vartheta_3^2 \vartheta_4}, & v_2 &= -e^{-i\pi/4} \frac{1}{\sqrt{\alpha_2}} \frac{\vartheta_4^3}{\vartheta_2 \vartheta_3^2}, \end{aligned} \tag{9.17}$$

where every theta constant in (9.17) is evaluated at $\tau - \tau_0$.

The roles of the parameters are now transparent. The constant τ_0 translates the modular solution and is fixed by the initial modal powers. The parameter a_0 controls the opposite phase

contributions in the two modes but does not alter their powers on the invariant manifold (9.9). The parameter a_2 weights the two quartic combinations in the Hamiltonian and, on $H = 0$, fixes the product $\alpha_1\alpha_2$ through (9.16); the separate values of α_1 and α_2 determine the distribution between each mode and its Hamiltonian conjugate.

Figure 2 compares (9.17) with an independent numerical integration of the original Hamilton equations, before making the $H = 0$ simplification used in (9.6). The elliptic-integral argument in the numerical system is calculated directly from the evolving modal powers, rather than from the modular lambda function. The numerical values satisfy $\alpha_1\alpha_2 = \coth(a_2/2) = 2$ and therefore select this same component of $H = 0$.

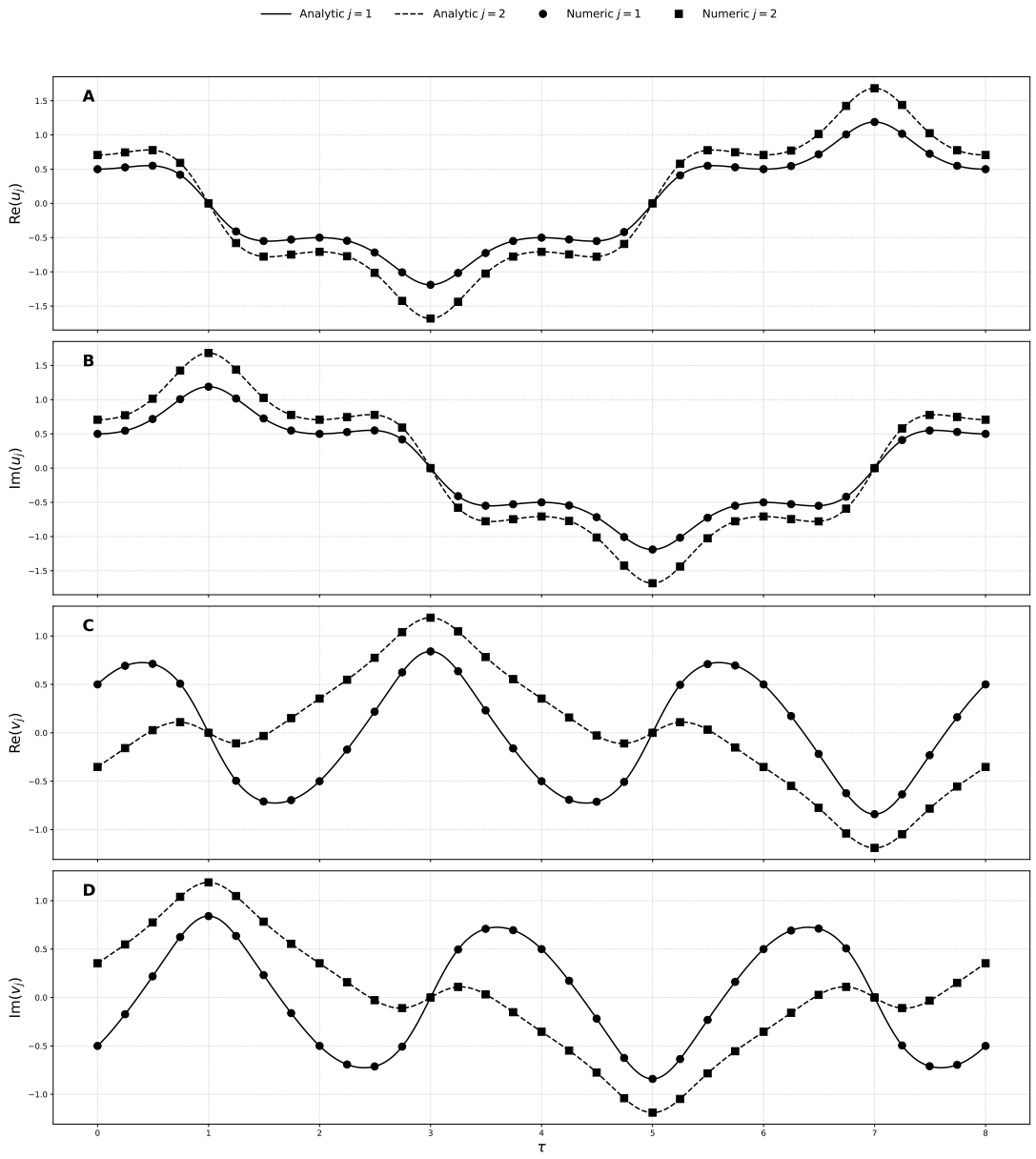


Figure 2: Comparison of the analytic theta-constant solutions (lines) with direct numerical integration of the original Hamilton equations (markers). The parameters are $a_0 = 1/2$, $a_2 = \log 3$, $\alpha_1 = 1$, $\alpha_2 = 2$, and $\tau_0 = -i$, so that $\alpha_1\alpha_2 = \coth(a_2/2)$. Numerical integration uses SciPy `solve_ivp` with DOP853, relative tolerance 10^{-10} and absolute tolerance 10^{-12}.

Finally, the construction has a direct relation to the Darboux–Halphen system. With primes

denoting differentiation with respect to τ, we use the convention

$$h_1' = h_2h_3 - h_1(h_2 + h_3), \tag{9.18a}$$
$$h_2' = h_3h_1 - h_2(h_3 + h_1), \tag{9.18b}$$
$$h_3' = h_1h_2 - h_3(h_1 + h_2). \tag{9.18c}$$

A standard map from the modular lambda function to a Darboux–Halphen solution is [33]

$$h_1 = -\frac{1}{2}\frac{d}{d\tau}\log\left(\frac{\lambda'}{\lambda}\right), \tag{9.19a}$$
$$h_2 = -\frac{1}{2}\frac{d}{d\tau}\log\left(\frac{\lambda'}{\lambda-1}\right), \tag{9.19b}$$
$$h_3 = -\frac{1}{2}\frac{d}{d\tau}\log\left(\frac{\lambda'}{\lambda(\lambda-1)}\right). \tag{9.19c}$$

Since (9.12) recovers both λ and $\lambda - 1$ directly from the modal powers, every solution of the present system on the selected invariant manifold determines this Darboux–Halphen solution. More directly, the map may be written entirely in terms of the modal powers and their ratio:

$$h_1 = -\frac{1}{2}\frac{d}{d\tau}\log\left[\frac{d}{d\tau}\log(u_1v_1)\right], \tag{9.20a}$$
$$h_2 = -\frac{1}{2}\frac{d}{d\tau}\log\left[\frac{d}{d\tau}\log(u_2v_2)\right], \tag{9.20b}$$
$$h_3 = -\frac{1}{2}\frac{d}{d\tau}\log\left[\frac{d}{d\tau}\log\left(\frac{u_2v_2}{u_1v_1}\right)\right]. \tag{9.20c}$$

This form holds for every value of a_0, since a_0 changes the ratios u_j/v_j but not the modal powers u_jv_j. In particular, when $a_0 = 0$, equation (9.14) gives $u_j/v_j = \alpha_j$. There is then no nontrivial phase evolution and $v_j = \alpha_j^{-1}u_j$, so the distinction between each mode and its Hamiltonian conjugate reduces to a constant rescaling. The Darboux–Halphen connection is therefore inherited from the generating equation; the focus here is the new canonical two-mode system and its exact theta-constant solutions.

10 Discussion

The construction developed here uses the functional freedom of an established Kummer-type Hamiltonian family to pass from prescribed scalar dynamics to coupled canonical systems. The intermodal power laws express every modal product as $u_jv_j = \gamma_j - \beta_j\rho$, so a single autonomous equation determines the power evolution and the remaining mode dependence is recovered by quadratures. Reading this reduction in reverse gives a family of Hamiltonians associated with a chosen scalar equation, or with a locally invertible function defining that equation. The examples demonstrate how this viewpoint both connects known systems and generates explicit systems across several classes of scalar dynamics.

A scalar equation does not uniquely determine the individual modes. For a fixed generating equation, the relation between P and Q leaves functional freedom that can change the mode and phase evolution while preserving the prescribed modal powers. Some apparent differences can be accounted for by transformations already available within the construction. In particular, Lemma 3.1 shows that, on $H = 0$ and where $P \neq 0$, the Hamiltonians H and H/P trace the same mode curves with different parametrisations. This does not remove the freedom associated with prescribing the scalar dynamics in the original variable z: the ratio Q/P can still depend on the choice of P, and the change of independent variable also depends on that choice. The total-modal-phase formulation gives a further relation between reconstructed modes

through transformations obtained by quadrature. A natural classification problem is therefore to determine when different choices of Hamiltonian functions produce systems related by canonical transformations, time reparameterisation, or the solution transformations described here, and which features distinguish the remaining choices.

The examples also distinguish the algebraic form of the mode equations from that of their scalar reduction. Cubic and quartic scalar polynomials give elliptic curves when nonsingular, while higher-degree squarefree polynomials lead to hyperelliptic curves. Repeated factors require normalization before the genus can be inferred, as the one-mode Fermat family illustrates; its incomplete-beta parametrisation also shows that the curve and the differential defining the independent variable must both be considered. By contrast, the Lotka–Volterra model and the other log-augmented examples show that polynomial mode equations can retain logarithmic terms in their scalar reductions. Thus polynomiality of the coupled equations does not by itself imply elliptic or hyperelliptic dynamics. A useful next step would be to identify conditions on P and Q that make the mode equations polynomial or rational, and to classify the scalar relations compatible with those conditions.

The passage from local quadratures to global mode solutions raises a separate set of questions. Even when ρ is meromorphic, reconstructing a mode by exponentiating an integral of its logarithmic derivative can introduce branching or nontrivial continuation around closed paths. For higher-genus scalar curves, inversion of the scalar Abelian integral brings additional constraints and need not define a globally single-valued meromorphic function of z. It would be valuable to formulate criteria, in terms of the singularities and periods of the reconstruction differentials, for the modes to be single-valued on a specified domain or meromorphic on an appropriate covering surface. Such criteria should also clarify continuation through zeros of modal products, where the logarithmic form of the reconstruction equations can become singular even when the original mode equations remain regular.

The non-polynomial examples suggest further constructive questions. The inverse spherical Bessel example starts directly from a locally invertible special function, while the modular λ example yields theta-constant mode solutions and a direct map to the Darboux–Halphen system. These cases invite a more systematic study of which generating functions permit the reconstruction quadratures to be evaluated within a specified class of special functions. For algebraic scalar reductions, the determinant identities discussed in the elliptic and hyperelliptic settings suggest investigating how the conserved mode products are encoded by identities on the associated curves. For nonalgebraic reductions, an analogous description would require structures adapted to the generating functions rather than an assumed algebraic curve.

These questions concern both the scalar equation and its canonical realisations. The present construction is local and applies on a selected Hamiltonian level and fixed intermodal invariants, with compatible choices of branches. Extending it therefore involves understanding how those local realisations continue, how they vary with the conserved quantities, and which equivalences preserve their reconstruction data. This provides a mathematical programme for organising the functional freedom of the Hamiltonian family beyond a list of explicitly solvable examples.

A Liouville integrability on the regular locus

We have already shown in Section 2 that the differential equations for ρ, u_j, v_j are separable and thus the system is solvable by quadratures. For completeness, this appendix verifies the commuting-integral criterion for Liouville integrability on the regular locus. Work on an open phase-space domain where the Hamiltonian is holomorphic, the modal products are nonzero, and at least one weight is nonzero.

From (2.11) we see that the intermodal power conservation laws may be written as:

$$B_{j,k} = \beta_k u_j v_j - \beta_j u_k v_k = \beta_k \gamma_j - \beta_j \gamma_k \tag{A.1}$$

for $j,k = 1 \dots N$, where the right hand side is constant. However, not all $B_{j,k}$ are independent. If, without loss of generality, we take $\beta_N \neq 0$, we may define $N-1$ intermodal power laws as:

$$C_l = \beta_N u_l v_l - \beta_l u_N v_N = \beta_N \gamma_l - \beta_l \gamma_N \tag{A.2}$$

for $l = 1, \dots, N-1$, then all other $B_{j,k}$ can be written as linear combinations of these via:

$$B_{j,k} = \frac{\beta_k}{\beta_N} C_j - \frac{\beta_j}{\beta_N} C_k, \tag{A.3}$$

for $j,k = 1 \dots N$ with the convention that $C_N = 0$. On this domain the differentials dx_m are independent: each is nonzero and supported on its own canonical pair (u_m, v_m). Holding x_N fixed, varying x_l changes $C_l = \beta_N x_l - \beta_l x_N$ and leaves the other C_m unchanged. Since $\beta_N \neq 0$, the $C_l, l = 1, \dots, N-1$, are independent.

The Hamiltonian is conserved because it has no explicit z dependence. Its functional independence from the C_l must be imposed on the locus under consideration:

$$dH \wedge dC_1 \wedge \cdots \wedge dC_{N-1} \neq 0. \tag{A.4}$$

A sufficient condition is $P(U-V) \neq 0$: varying a pair with $\beta_j \neq 0$ by $u_j \mapsto u_j e^{\theta}$, $v_j \mapsto v_j e^{-\theta}$ fixes every x_m and gives $dH/d\theta = \beta_j P(U-V)$ at $\theta = 0$. Other points may also satisfy (A.4); no independence claim is made on degenerate loci or for coefficient choices that make H dependent on the C_l.

We will now show that the $C_l, l = 1, \dots, N-1$ and H are in involution. We use the canonical Poisson bracket

$$\{F, G\} = \sum_{j=1}^{N} \left(\partial_{u_j} F \, \partial_{v_j} G - \partial_{v_j} F \, \partial_{u_j} G \right). \tag{A.5}$$

The modal powers $x_j = u_j v_j$ are mutually in involution, since, for all j,k:

$$\{x_j, x_k\} = 0, \tag{A.6}$$

and, since each C_l is a linear combination of modal powers, it follows immediately that

$$\{C_l, C_m\} = 0. \tag{A.7}$$

Moreover, Hamilton's equations give

$$\{x_j, H\} = \frac{dx_j}{dz} = -\beta_j P \left(\prod_{r=1}^{N} u_r^{\beta_r} - \prod_{r=1}^{N} v_r^{\beta_r} \right), \tag{A.8}$$

therefore:

$$\{C_l, H\} = \beta_N \{x_l, H\} - \beta_l \{x_N, H\} = 0. \tag{A.9}$$

Thus $C_1, \dots, C_{N-1}, H$ are in involution and are independent on the locus (A.4), establishing local Liouville integrability there. For a real symplectic restriction, the Liouville–Arnold torus and action–angle conclusion additionally requires a compact connected regular common level set; these global hypotheses are not asserted for the general complex systems considered here.

B Weierstrass form under reparameterisation

Consider the logarithmically deformed Weierstrass equation (6.16), and write

$$W(s) = 4s^3 - g_2 s - g_3. \tag{B.1}$$

Let $w(h)$ solve the ordinary Weierstrass equation

$$\left(\frac{dw}{dh}\right)^2 = W(w), \tag{B.2}$$

so that, after choosing a shift h_*,

$$w(h) = \wp(h - h_*; g_2, g_3). \tag{B.3}$$

We seek a solution of the deformed equation in the form

$$s(z) = w(h(z)). \tag{B.4}$$

Substitution into (6.16) gives

$$\left(\frac{dw}{dh}\right)^2 h'^2 = \left(\frac{dw}{dh}\right)^2 + c_5(w + c_6)^2 \log^2\left(\frac{4(w + c_6)}{c_0}\right). \tag{B.5}$$

Now define

$$L(h) = \log\left[\log\left(\frac{4(w(h) + c_6)}{c_0}\right)\right]. \tag{B.6}$$

Its derivative is

$$\frac{dL}{dh} = \frac{dw/dh}{(w + c_6)\log\left(4(w + c_6)/c_0\right)}. \tag{B.7}$$

It follows that the equation for the new variable h is

$$h'^2 = 1 + \frac{c_5}{(dL/dh)^2}. \tag{B.8}$$

Let $h_0 = h(z_0)$ and define, locally on consistent branches,

$$\mathcal{Z}(h) = \int_{h_0}^{h} \frac{L'(\tau)}{\sqrt{L'(\tau)^2 + c_5}}\, d\tau. \tag{B.9}$$

Equation (B.8) is then equivalent to

$$z - z_0 = \mathcal{Z}(h(z)). \tag{B.10}$$

Where $\mathcal{Z}$ is locally invertible, the solution may therefore be written

$$s(z) = \wp\left(\mathcal{Z}^{-1}(z - z_0) - h_*; g_2, g_3\right). \tag{B.11}$$

There is also a simple limiting form when the logarithmic deformation is dominant. If

$$|c_5| \gg |L'(h)|^2 \tag{B.12}$$

along the part of the solution under consideration, then, on a chosen branch of $\sqrt{c_5}$,

$$\mathcal{Z}(h) \sim \frac{1}{\sqrt{c_5}} \int_{h_0}^{h} L'(\tau)\, d\tau = \frac{L(h) - L(h_0)}{\sqrt{c_5}}. \tag{B.13}$$

Consequently,

$$L(h(z)) \sim L(h_0) + \sqrt{c_5}(z - z_0), \qquad h(z) \sim L^{-1}\left(L(h_0) + \sqrt{c_5}(z - z_0)\right). \tag{B.14}$$

Thus, in this regime, the reparameterisation is approximately obtained by directly inverting the double-logarithmic function L.

When $c_5 = 0$, the integrand in (B.9) is locally ± 1, so h differs from z only by a sign and an additive constant and the ordinary Weierstrass solution is recovered. When $c_5 \neq 0$, the dependence on the Weierstrass function remains, but the relation between h and z requires inversion of the integral (B.9). Thus the logarithmically deformed equation is elliptic in the reparameterised variable h, but is not generally an elliptic function of z.

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